Exercises

12.4.1 Verify that the function $f(x,y) = \sqrt{|xy|}$ is continuous at $(0,0)$.

12.4.2 Verify that for $h_1, \ldots, h_n$ real numbers

$$\sqrt{h_1^2 + \cdots + h_n^2} \leq |h_1| + \cdots + |h_n| \leq \sqrt{n}\sqrt{h_1^2 + \cdots + h_n^2}.$$

Use these inequalities to obtain another (equivalent) definition for *differentiability* of a function of n variables at a point $\mathbf{x_0}$ that indicates that $f(\mathbf{x}) - f(\mathbf{x_0})$ can be approximated near $\mathbf{x_0}$ by a linear function T in such a way that $|f(x) - T(x)| \to 0$ as $\|x - x_0\|_1 \to 0$.

12.4.3 For $h = (h_1, \ldots, h_n) \in \mathbb{R}^n$ write

$$\begin{aligned} \|h\|_1 &= |h_1| + \cdots + |h_n| \\ \|h\|_2 &= \|h\| = \sqrt{h_1^2 + \cdots + h_n^2} \\ \|h\|_\infty &= \max(|h_1|, \ldots, |h_n|). \end{aligned}$$

Show that the following four conditions are equivalent:

$$h_i \to 0 \text{ for all } i = 1, \ldots, n,$$
$$\|\mathbf{h}\|_1 \to 0, \quad \|\mathbf{h}\|_2 \to 0, \quad \|\mathbf{h}\|_\infty \to 0$$

Provide yet another definition of *differentiability* using $\|\mathbf{h}\|_\infty$ in place of $\|\mathbf{h}\|_1$ or $\|h\|_2$.

See Note 277

12.4.4 Show that

$$f(x,y) = \begin{cases} \frac{x^3 - xy^2}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0) \end{cases}$$

is continuous and has first-order partial derivatives on $\mathbb{R}^2$, but is not differentiable at $(0,0)$.

12.4.5 Some authors avoid the use of partial derivatives in defining differentiability. Definition 12.16 then takes the following form.

Definition. Let f be defined on an open set $D \subset \mathbb{R}^n$ and let

$$x = (x_1, \ldots, x_n) \in D.$$

Then f is differentiable at x if there exists a linear function $L : \mathbb{R}^n \to \mathbb{R}$ such that when ε is defined by

$$f(x_1 + h_1, \ldots, x_n + h_n) - f(x_1, \ldots, x_n) = \sum_{i=1}^{n} L(x)h_i + \varepsilon\sqrt{(|h_1|^2 + \cdots + |h_n|^2)}$$

with $\varepsilon(0) = 0$, we have $\varepsilon \to 0$ when $\sqrt{(|h_1|^2 + \cdots + |h_n|^2)} \to 0$. Prove that this definition is equivalent to Definition 12.16.

See Note 278

12.4.3 Differentiability and Continuity

We now show that differentiable functions are continuous. We first consider the case $n = 2$. We do this because the essentials of a proof are already contained in the special case $n = 2$, while the notation is simpler in $\mathbb{R}^2$. The relevant pictures that go with the proof are easier to visualize.

Theorem 12.17: *If f is differentiable at (x_0,y_0), then f is continuous at (x_0,y_0).*

Proof. We must prove that

$$\lim_{(h,k)\to(0,0)} f(x_0+h,y_0+k) = f(x_0,y_0).$$

Since f is differentiable at (x_0,y_0), we can write

$$f(x_0+h,y_0+k) - f(x_0,y_0) = f_1(x_0,y_0)h + f_2(x_0,y_0)k + \varepsilon(|h|+|k|),$$

where $\varepsilon \to 0$ as $(h,k) \to (0,0)$. We obtain from the triangle inequality that

$$\begin{aligned} &|f(x_0+h,y_0+k) - f(x_0,y_0)| \leq \\ &\quad |f_1(x_0,y_0)||h| + |f_2(x_0,y_0)||k| + |\varepsilon|(|h|+|k|) \end{aligned} \tag{28}$$

Now $f_1(x_0,y_0)$ and $f_2(x_0,y_0)$ are fixed numbers, so the right side of the inequality (28) approaches 0 as $(h,k) \to (0,0)$. Thus the left side also approaches 0. But that is an equivalent formulation of the requirement

$$\lim_{(h,k)\to(0,0)} f(x_0+h,y_0+k) = f(x_0,y_0).$$

■

Note. Observe that we needed only finiteness of f_1 and f_2 at (x_0,y_0) and that $\varepsilon(h,k)(|h|+|k|) \to 0$ as $(h,k) \to (0,0)$ to infer that the right side of (28) approaches 0 as $(h,k) \to (0,0)$. Differentiability gave us more than we needed, namely that $\varepsilon(h,k) \to 0$ as $(h,k) \to (0,0)$. (See Exercise 12.4.7.)

The proof of the generalization to $\mathbb{R}^n$ is essentially the same. We leave this as Exercise 12.4.6.

Exercises

12.4.6 Prove that if f is defined in a neighborhood of a point $x \in \mathbb{R}^n$ and differentiable at x, then f is continuous at x.

12.4.7 Give an example of a function $f : \mathbb{R}^2 \to \mathbb{R}$ such that

$$|f(h,k) - f(0,0)| = f_1(0,0)h + f_2(0,0)k + \varepsilon(|h|+|k|)$$

where $\varepsilon(|h|+|k|) \to 0$ as $(h,k) \to (0,0)$ but $\varepsilon \not\to 0$ as $(h,k) \to (0,0)$.

12.4.8 Give an example of a differentiable function of two variables whose partials are discontinuous at $(0,0)$.

12.4.4 Directional Derivatives

In Example 12.15 we saw that the function $f(x,y) = \sqrt{|xy|}$ had partial derivatives equal to 0 at $(0,0)$, but was not differentiable. We identified a problem in one direction. The function does not have a directional derivative in that direction (Exercise 12.4.11). We next see that a differentiable function has directional derivatives in all directions and that, moreover, all directional derivatives may be computed from the partials by a simple formula.

Theorem 12.18: *Let f be differentiable at $(x_0,y_0)\in\mathbb{R}^2$ and let $\mathbf{u}=(u_1,u_2)$ be any unit vector. Then $D_{\mathbf{u}}f(x_0,y_0)$ exists and*

$$D_{\mathbf{u}}f(x_0,y_0)=f_1(x_0,y_0)u_1+f_2(x_0,y_0)u_2. \tag{29}$$

Proof. Since f is differentiable at (x_0,y_0) we can write

$$\begin{aligned}f(x_0+h,y_0+k)-f(x_0,y_0)=\\ f_1(x_0,y_0)h+f_2(x_0,y_0)k+\varepsilon(|h|+|k|),\end{aligned}$$

where $\varepsilon\to 0$ as $(h,k)\to(0,0)$. Thus

$$\begin{aligned}f(x_0+tu_1,y_0+tu_2)-f(x_0,y_0)=\\ f_1(x_0,y_0)tu_1+f_2(x_0,y_0)tu_2+\varepsilon(|tu_1|+|tu_2|)\end{aligned} \tag{30}$$

and $\varepsilon\to 0$ as $t\to 0$.

Dividing both sides of (30) by t and letting $t\to 0$, we obtain

$$\begin{aligned}\lim_{t\to 0}\frac{f(x_0+tu_1,y_0+tu_2)-f(x_0,y_0)}{t}=\\ \lim_{t\to 0}(f_1(x_0,y_0)u_1+f_2(x_0,y_0)u_2+\varepsilon\frac{|t|}{t}(|u_1|+|u_2|)).\end{aligned}$$

Now, $|u_1|+|u_2|$ is a constant, $\left|\frac{|t|}{t}\right|=1$, and $\varepsilon\to 0$ as $t\to 0$, so the limit equals $f_1(x_0,y_0)u_1+f(x_0,y_0)u_2$ as required. ■

Theorem 12.18 is valid for functions on $\mathbb{R}^n$ for every n, the proof being similar. We leave this proof as Exercise 12.4.12.

The Gradient Observe that, for a differentiable function f, the identity

$$D_{\mathbf{u}}f(x_0,y_0)=f_1(x_0,y_0)u_1+f_2(x_0,y_0)u_2$$

can be written in the form

$$D_{\mathbf{u}}f(x_0,y_0)=(f_1(x_0,y_0),f_2(x_0,y_0))\cdot(u_1,u_2).$$

The vector (f_1,f_2) or $(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y})$, where the partials are evaluated at (x_0,y_0), is of sufficient importance to have a name. It is called the *gradient* of f at (x_0,y_0) and denoted by

$$\mathbf{grad}\ f(x_0,y_0)\quad\text{or}\quad\bigtriangledown f(x_0,y_0).$$

By the law of cosines (or Exercise 11.2.4) we can write

$$\begin{aligned}D_{\mathbf{u}}f(x_0,y_0)&=(\bigtriangledown f(x_0,y_0))\cdot u\\ &=\|\bigtriangledown f(x_0,y_0)\|\,\|u\|\cos\theta\\ &=\|\bigtriangledown f(x_0,y_0)\|\cos\theta\end{aligned}$$

where θ is the angle between the vectors $\bigtriangledown f(x_0,y_0)$ and u. [Here we must have $\bigtriangledown f(x_0,y_0)\neq 0$, otherwise θ is not defined.] Thus the maximum rate of change of f is in the direction corresponding to $\theta=0$ (where $\cos\theta$ achieves its maximum). This occurs when u and $\bigtriangledown f(x_0,y_0)$ have the same direction. The magnitude of the rate of increase of f in this direction is $\|\bigtriangledown f(x_0,y_0)\|$.

For reference we state this as a theorem, expressed for functions of several variables.

Theorem 12.19: *If $f : \mathbb{R}^n \to \mathbb{R}$ is a differentiable function on an open set D, then at each point x in D at which the gradient of f does not vanish the maximum rate of change of f is in the direction $\bigtriangledown f(x)$ and the magnitude of the rate of increase of f in this direction is $\|\bigtriangledown f(x)\|$.*

Exercises 12.4.17 and 12.4.18 show that the preceding discussion might fail if we weaken the assumption of differentiability to the mere existence of the partials at (x_0, y_0).

Exercises

12.4.9 What are the directional derivatives for a function $f : \mathbb{R}^n \to \mathbb{R}$ if $n = 1$?
SEE NOTE 279

12.4.10 Is it still true that the gradient is in the direction of greatest change for the function if $f : \mathbb{R}^n \to \mathbb{R}$ where $n = 1$?

12.4.11 Verify that the function $f(x,y) = \sqrt{|xy|}$ does not have a directional derivative in the direction $(\sqrt{2}/2, \sqrt{2}/2)$ at the point $(0,0)$.

12.4.12 State and prove the extension of Theorem 12.18 to functions of n variables.

12.4.13 Verify that the function

$$f(x,y) = \frac{x^2 y}{x^4 + y^2} \ , \ f(0,0) = 0$$

is not differentiable at $(0,0)$ and yet has directional derivatives in every direction.

12.4.5 An Example

We have already seen that a continuous function can possess partial derivatives at a point without being differentiable at that point. More remarkably, a continuous function can possess directional derivatives *in every direction* and still be nondifferentiable. Exercise 12.4.13 has exhibited a *discontinuous* function that possesses all directional derivatives; since it is not continuous it cannot be differentiable. In this section we shall describe geometrically a *continuous* function with these properties. The picture we'll describe should be instructive, while a computational analysis would be a bit tedious. (See Exercise 12.4.14.)

Example 12.20: We build the function f in stages. First we let $f(x,0) = x$ for all $x \in \mathbb{R}$. This will guarantee that $f_1(0,0) = 1$. Next, for $|y| \geq x^2$, we define $f(x,y) = 0$. This will guarantee that all other directional derivatives are 0 at $(0,0)$.

To see this, we need only observe that any line through $(0,0)$, other than the x-axis, intersects the set

$$\{(x,y) : |y| \geq x^2\}$$

on a line segment containing $(0,0)$.

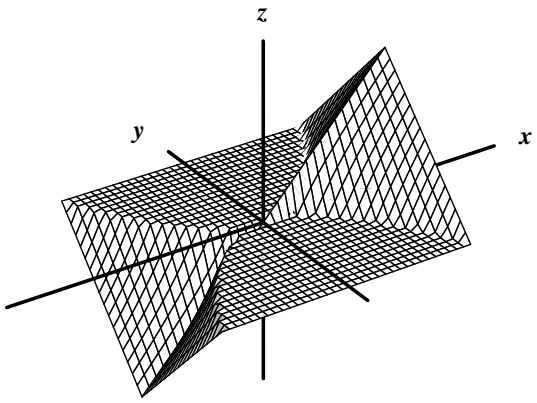


Figure 12.5. A continuous, nondifferentiable function with directional derivatives in every direction.

Finally, on the set

$$\{(x,y) : |y| < x^2\}$$

we define f in any manner that makes f continuous on $\mathbb{R}^2$. For example, f can be linear on each vertical segment joining a point of the parabola $y = x^2$ or $y = -x^2$ to the x-axis, with the value 0 on the parabola and x on the x-axis. Figure 12.5 illustrates.

Thus f has directional derivatives 0 in all directions except for

$$(u_1, u_2) = (1, 0)$$

or

$$(u_1, u_2) = (-1, 0).$$

If f were differentiable at $(0,0)$, we would have, by Theorem 12.18, that for any unit vector $\mathbf{u} = (u_1, u_2)$,

$$D_{\mathbf{u}} f(0,0) = f_1(0,0)u_1 + f_2(0,0)u_2 = 1 \cdot u_1 + 0 \cdot u_2 = u_1.$$

But we have seen that $D_{\mathbf{u}} f(0,0) = 0$ unless $u_2 = 0$, so the equality

$$D_{\mathbf{u}} f(0,0) = u_1$$

is valid only for $u_1 = 0$ or $u_2 = 0$. Since Theorem 12.18 does not give the correct directional derivatives, f is not differentiable at $(0,0)$. ◀

Observe that for the function we described, all directional derivatives except the ones in the direction of the positive or negative x-axis require the tangent plane to be the plane $z = 0$. But the tangent line that lies in the plane $y = 0$ does not lie in this plane. In short, there can be no tangent plane. See also Exercise 12.4.15.

Example 12.20 also shows that Theorem 12.18 is not valid if we drop the requirement that f be differentiable. Directional derivatives cannot always be computed from formula (29) if that assumption is dropped. (But see Exercise 12.4.15.)

Exercises

12.4.14 An analytic representation of the example we gave in the text (Example 12.20) can take the form

$$f(x) = \begin{cases} 0, & \text{if } |y| \geq x^2 \\ x, & \text{if } y = 0 \\ -\frac{1}{x}(y - x^2), & \text{if } 0 < y < x^2 \\ \frac{1}{x}(y + x^2), & \text{if } -x^2 < y < 0 \end{cases}$$

(a) Verify analytically that f is continuous and has directional derivative $D_u f(0,0) = 0$, for all directions with the exception of $(1,0)$ and $(-1,0)$.

(b) Calculate the partial derivatives f_1 and f_2, and show they are discontinuous at $(0,0)$. [That f_1 and f_2 are discontinuous at $(0,0)$ follows also from Theorem 12.21, proved later.]

12.4.15 A careless student states "The functions in Example 12.20 and Ex. 12.4.14 fail to be differentiable at $(0,0)$ even though all derivatives exist at $(0,0)$ because the tangent lines at $(0,0)$ don't all lie in the same plane, so there is no tangent plane. If all tangent lines at a point exist and *do* lie in the same plane, then that plane must be the tangent plane." Is the second statement correct? That is, if $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous on $\mathbb{R}^2$ with $f(0,0) = 0$ and *every* directional derivative at $(0,0)$ is zero, then f is differentiable at $(0,0)$ and the xy-plane is the tangent plane at $(0,0,0)$.

SEE NOTE 280

12.4.6 Sufficient Conditions for Differentiability

It is not always easy to prove that a function is differentiable directly from Definition 12.16. The same was true in dealing with functions of one variable. There we obtained general theorems that often simplified our task. We next present a theorem that can often be applied to show differentiability for functions of several variables.

Theorem 12.21: *Let f be defined in a neighborhood D of (x_0, y_0). Suppose one of the partial derivatives is defined on D and is continuous at (x_0, y_0), while the other partial is defined at least at (x_0, y_0) (and finite there). Then f is differentiable at (x_0, y_0).*

Proof. Suppose f_1 is continuous at (x_0, y_0), the proof being similar if it is f_2 that is continuous there. We wish to show that we can write

$$\begin{aligned} f(x_0 + h, y_0 + k) - f(x_0, y_0) = \\ f_1(x_0, y_0)h + f_2(x_0, y_0)k + \varepsilon(|h| + |k|), \end{aligned} \tag{31}$$

where $\varepsilon \to 0$ as $(h,k) \to 0$.

We shall express the left side of (31) as a sum of two terms. We apply the mean value theorem to one of the terms and use the continuity of f_1 at (x_0, y_0) to obtain an estimate of the first term on the right side of (31). We then estimate the second term using only the existence of f_2 at (x_0, y_0). Manipulating these estimates will give the desired result.

To begin, write

$$\begin{aligned} f(x_0 + h, y_0 + k) - f(x_0, y_0) = \\ [f(x_0 + h, y_0 + k) - f(x_0, y_0 + k)] + [f(x_0, y_0 + k) - f(x_0, y_0)]. \end{aligned} \tag{32}$$

Applying the mean value theorem (Theorem 7.20) to the first bracketed term, we obtain a number x' between x_0 and x_0+h such that

$$f(x_0+h,y_0+k)-f(x_0,y_0+k)=f_1(x',y_0+k)h. \tag{33}$$

Now f_1 is continuous at (x_0,y_0) so

$$\lim_{x'\to x_0,k\to 0} f_1(x',y_0+k)=f_1(x_0,y_0).$$

Thus we can write

$$f_1(x',y_0+k)=f_1(x_0,y_0)+\varepsilon_1 \tag{34}$$

where $\varepsilon_1\to 0$ as $x'\to x_0$ and $k\to 0$. Note that $x'\to x_0$ as $h\to 0$, since x' is between x_0 and x_0+h.

We now consider the second bracketed term. Since f_2 is finite at (x_0,y_0), we can write for $k\neq 0$

$$\frac{f(x_0,y_0+k)-f(x_0,y_0)}{k}=f_2(x_0,y_0)+\varepsilon_2$$

where $\varepsilon_2\to 0$ as $k\to 0$. This we can write in the form

$$f(x_0,y_0+k)-f(x_0,y_0)=f_2(x_0,y_0)k+\varepsilon_2 k. \tag{35}$$

Substituting (33) and (35) into (32) and then using (34), we arrive at the equality

$$\begin{aligned} &f(x_0+h,y_0+k)-f(x_0,y_0)= \\ &\quad f_1(x_0,y_0)h+f_2(x_0,y_0)k+(\varepsilon_1 h+\varepsilon_2 k) \end{aligned} \tag{36}$$

where both ε_1 and ε_2 approach 0 as $(h,k)\to(0,0)$.

Comparing (36) with the desired form (31), we see that we must replace the term $(\varepsilon_1 h+\varepsilon_2 k)$ by a term of the form $\varepsilon(|h|+|k|)$. Letting

$$\varepsilon=\frac{\varepsilon_1 h+\varepsilon_2 k}{|h|+k|}$$

does the job. Indeed, with this value for ε,

$$\varepsilon_1 h+\varepsilon_2 k=\varepsilon(|h|+|k|),$$

so (36) reduces to (31). Furthermore, $\varepsilon\to 0$ as $(h,k)\to(0,0)$. To see this observe that

$$|\varepsilon|=\left|\frac{\varepsilon_1 h}{|h|+|k|}+\frac{\varepsilon_2 k}{|h|+|k|}\right|\leq\left|\frac{\varepsilon_1 h}{|h|+|k|}\right|+\left|\frac{\varepsilon_2 k}{|h|+|k|}\right|\leq|\varepsilon_1|+|\varepsilon_2|.$$

Since $\varepsilon_1\to 0$ and $\varepsilon_2\to 0$ as $(h,k)\to(0,0)$, it follows that $\varepsilon\to 0$. ■

The conditions of Theorem 12.21 are often met and may be easier to verify than verifying differentiability directly from the definition. As before, the general case for functions of n variables can be obtained with a similar proof (Exercise 12.4.16).

Exercises

12.4.16 State and prove the extension of Theorem 12.21 to functions of n variables.
SEE NOTE 281

12.4.17 Consider a function f constructed so as to be continuous and such that

(a) $f(x,y) = 0$ unless $x > 0$ and $x^2 < y < 3x^2$,

(b) for each $x > 0$, $f(x, 2x^2) = x$, and

(c) $0 \le f(x,y) \le x$ for all (x,y) with $x > 0$.

Then all directional derivatives vanish at $(0,0)$ (this was given in the hint to Exercise 12.4.15). Show that this function has no direction of maximum change at $(0,0)$. Modify this example to obtain a function g with

$$g_1(0,0) = 1 = g_2(0,0)$$

yet there is no direction of maximum change.

12.4.18 Give an example of a continuous function $f : \mathbb{R}^2 \to \mathbb{R}$ having partial derivatives at $(0,0)$ with

$$f_1(0,0) \neq 0 \ , \ f_2(0,0) \neq 0$$

but the vector $(f_1(0,0), f_2(0,0))$ does not point in the direction of maximal change, even though there is such a direction.

12.4.7 The Differential

Suppose f is differentiable at a point $(x_0, y_0) \in \mathbb{R}^2$. Then we can write

$$\begin{aligned} &f(x_0+h, y_0+k) - f(x_0,y_0) = \\ &\quad f_1(x_0,y_0)h + f_2(x_0,y_0)k + \varepsilon(|h|+|k|) \end{aligned} \tag{37}$$

where $\varepsilon \to 0$ as $(h,k) \to (0,0)$.

Let us rewrite this in a form that may be more familiar from elementary calculus. Let $z = f(x,y)$. Write $h = \triangle x$, $k = \triangle y$, $\triangle f = \triangle z$. Then

$$\triangle f = \triangle z = f(x_0 + \triangle x, y_0 + \triangle y) - f(x_0, y_0)$$

represents the change in f (or z) that corresponds to a change in x and y given by $\triangle x$ and $\triangle y$. With this notation, (37) becomes

$$\triangle f = f_1(x_0,y_0)\triangle x + f_2(x_0,y_0)\triangle y + \varepsilon(|\triangle x| + |\triangle y|). \tag{38}$$

The term $\varepsilon(|\triangle x| + |\triangle y|)$ represents the error in estimating $\triangle z$, the change in z, by the change in the tangent plane at (x_0, y_0) corresponding to changes of $\triangle x$ and $\triangle y$, respectively. It has been customary historically, when obvious limits are involved, to use notation such as

$$df = f_1 dx + f_2 dy \ \text{ or } \ dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} y.$$

What do we mean by such notation? To be precise, df is a function depending on x, y, dx, and dy:

$$df = df(x,y,dx,dy).$$

(So x, y, dx, and dy are independent variables and df is a function of these four variables.) This function df is of sufficient importance to deserve a name.

Definition 12.22: Let f be differentiable at a point (x,y). The function df defined by

$$df(x,y,dx,dy) = f_1 dx + f_2 dy$$

is called the *differential of f at (x,y)*.

Let's look at a simple example to illustrate the concepts.

Example 12.23: Let $f(x,y) = x^2y^3$, $x = 3$, $y = 1$, $\triangle x = \triangle y = .01$. We'll compute $\triangle f$ and df, the approximation to $\triangle f$. Here $x + \triangle x = 3.01$ and $y + \triangle y = 1.01$, so

$$\triangle f = f(3.01, 1.01) - f(3,1) = 9.3346 - 9 = .3346$$

to four decimal places. Our approximation, $df(3,1,.01,.01)$, becomes

$$df = f_1(3,1)\,dx + f_2(3,1)\,dy = (6 \times .01) + (27 \times .01) = .33.$$

Thus the error in using the differential df for estimating the actual change $\triangle f$ is only about .0046. Note that this error is small in comparison with $|\triangle x| + |\triangle y| = .02$. ◀

Functions of Several Variables When dealing with more than two variables, the situation is similar.

Definition 12.24: Let f be differentiable at $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$. The *differential of f at x* is the function of $2n$ variables given by

$$df(x_1, \ldots, x_n, h_1, \ldots, h_n) = f_1 h_1 + \cdots + f_n h_n \tag{39}$$

As before, it is sometimes convenient to write (39) in the form

$$df = \frac{\partial f}{\partial x_1} dx_1 + \cdots + \frac{\partial f}{\partial x_n} dx_n. \tag{40}$$

As is often true with alternative notations when dealing with derivatives, the notation (40) is suggestive of various formulas. For example, if u and v are differentiable at a point $x \in \mathbb{R}^n$, then

$$\begin{aligned} d(u+v) &= du + dv\,,\ d(uv) = u\,dv + v\,du\,, \\ d\left(\frac{u}{v}\right) &= \frac{v\,du - u\,dv}{v^2} \quad (v \neq 0). \end{aligned} \tag{41}$$

To check the product formula, for example, note that by definition each of the differentials can be written as

$$\begin{aligned} du &= \frac{\partial u}{\partial x_1} dx_1 + \cdots + \frac{\partial u}{\partial x_n} dx_n, \\ dv &= \frac{\partial v}{\partial x_1} dx_1 + \cdots + \frac{\partial v}{\partial x_n} dx_n, \\ d(uv) &= \frac{\partial (uv)}{\partial x_1} dx_1 + \cdots + \frac{\partial (uv)}{\partial x_n} dx_n. \end{aligned} \tag{42}$$

But the product rule clearly is valid for partial derivatives. Thus (42) becomes

$$\begin{aligned} d(uv) = \left[u\frac{\partial v}{\partial x_1} + v\frac{\partial u}{\partial x_1}\right] dx_1 + \cdots + \left[u\frac{\partial v}{\partial x_n} + v\frac{\partial u}{\partial x_n}\right] dx_n = \\ u\left[\frac{\partial v}{\partial x_1} dx_1 + \cdots + \frac{\partial v}{\partial x_n} dx_n\right] + \cdots + v\left[\frac{\partial u}{\partial x_1} dx_1 + \cdots + \frac{\partial u}{\partial x_n} dx_n\right] \\ = u\,dv + v\,du. \end{aligned}$$

Similarly, differentials of elementary functions are as expected as the example now shows.

Example 12.25: If $u : \mathbb{R}^n \to \mathbb{R}$ is differentiable and we are allowed, for the moment, to assume that e^u is also differentiable, then

$$\begin{aligned} d(e^u) &= \frac{\partial e^u}{\partial x_1}dx_1 + \cdots + \frac{\partial e^u}{\partial x_n}dx_n \\ &= e^u \frac{\partial u}{\partial x_1}dx_1 + \cdots + e^u \frac{\partial u}{\partial x_n}dx_n \\ &= e^u \left[\frac{\partial u}{\partial x_1}dx_1 + \cdots + \frac{\partial u}{\partial x_n}dx_n \right] \\ &= e^u du \end{aligned}$$

thus assuming a familiar form. (See Exercise 12.8.7 for a discussion of whether e^u is differentiable.) ◀

Exercises

12.4.19 Calculate $\triangle f$ and df in Example 12.23 when $\triangle x = .001$ and $\triangle y = .002$. Compare the resulting error with $|\triangle x| + |\triangle y| = .003$.

12.4.20 Verify the formulas in (41) for $d(u+v)$ and $d\left(\frac{u}{v}\right)$.

12.4.21 Since the definition of differential involves differentiability of the functions, the formulas (41) require the differentiability of the functions u and v, uv and u/v. Prove that when u and v are differentiable at a point $(x_0, y_0) \in \mathbb{R}^2$, then so too are their sum, product, and quotient. [For the quotient assume also that $v(x_0, y_0) \neq 0$].

12.5 Chain Rules

We saw in Example 12.25 that if we have a formula such as $z = e^u$, where u is a real-valued function of several variables, we can compute its differential $dz = e^u du$, as we would for functions of one real variable. We can view this as an instance of a *chain rule*. Actually there are many chain rules. They involve computing differentials or partial derivatives of functions defined from other functions via composition. We discuss such situations in this section and show how a chain is created and what the resulting chain rule should be. We also give an indication of why the chain rule should work, and then proceed to a formal proof.

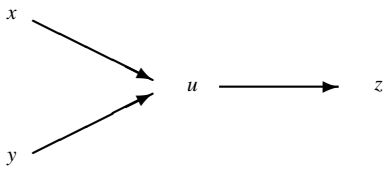
12.5.1 Preliminary Discussion

We begin with three examples, discussing each of them informally. To keep this informal discussion simple, we avoid technicalities such as the domains of definition of the functions and the precise hypotheses needed for the resulting chain rules.

Example 12.26: Let $u = u(x, y)$ and let $z = F(u)$. We can view z as a function G of x and y via the intermediate variable u. Thus

$$z = F(u) = F(u(x, y)) = G(x, y).$$

The dependencies of the variables can be described as "z depend on u, and u depends on x and y" and expressed schematically:



(43)

This has associated chain rules

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} \text{ and } \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} \tag{44}$$

that you may remember from calculus.

Ultimately z depends on the two variables x and y via the single variable u. Thus there are two partials to compute, $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, each involving a one-term chain as seen in (44). ◀

Example 12.27: For a concrete example, let $z = \sin(x^2 + y^3) = G(x,y)$, and write $u(x,y) = x^2 + y^3$ and $F(u) = \sin u$. This corresponds to the schema in (43). We would then conclude, using the chain rule (44), that

$$\frac{\partial z}{\partial x} = 2x\cos(x^2+y^3) \text{ and } \frac{\partial z}{\partial y} = 3y^2\cos(x^2+y^3).$$

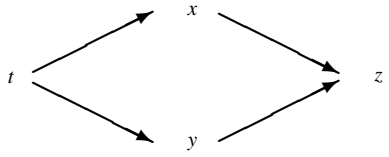
◀

Let us complicate things a bit.

Example 12.28: Let $x = f(t)$, $y = g(t)$, $z = F(x,y)$. Here we can view z as a function of t:

$$z = F(x,y) = F(f(t),g(t)) = G(t).$$

The dependencies of the variables can be described as "z depends on x and y, each of which depends on t" and expressed schematically:



with associated chain rule

$$G'(t) = \frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}. \tag{45}$$

Note that x and y are functions of t alone, as is G, so ordinary derivatives are involved. On the other hand, $F(x,y)$ is a function of two variables, so

$$\frac{\partial z}{\partial x} = F_1 \text{ and } \frac{\partial z}{\partial y} = F_2$$

are *partial* derivatives. In contrast with Example 12.26, there is now only one derivative we wish to calculate, $G'(t) = \frac{dz}{dt}$, but the chain rule involves *two* terms to be added, one term arising from each "path from t to z." ◀

Example 12.29: Again, for a concrete example, let

$$x = t^2,\ y = t^3,\ \text{and}\ \ z = xy.$$

Then from (45),

$$\frac{dz}{dt} = y2t + x3t^2 = 2t^4 + 3t^4 = 5t^4,$$

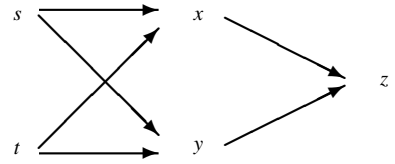
as expected when we observe that $G(t) = t^5$. ◀

Let's complicate matters a bit more.

Example 12.30: Let $z = F(x,y)$, $x = f(s,t)$, $y = g(s,t)$. We can view z as a function of s and t via x and y:

$$z = F(x(s,t),y(s,t)) = G(s,t).$$

The dependencies of the variables can be described as "z depends on both x and y, and each in turn depends on both s and t" and expressed schematically:



To compute $\frac{\partial z}{\partial s}$ we follow each path from s to z. Thus

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \tag{46}$$

and, similarly, following each path from t to z,

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}.$$

◀

Example 12.31: Let

$$z = F(x,y) = x^2 + y^3,$$

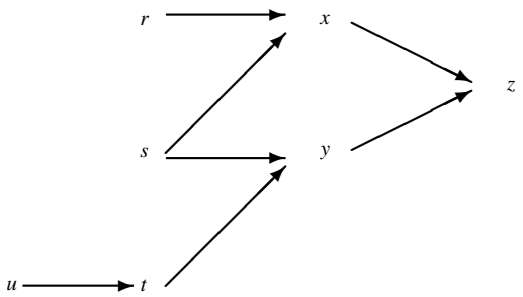
$x = f(s,t) = st$, and $y = g(s,t) = e^{st}$. Then the chain rule in (46) gives

$$\frac{\partial z}{\partial s} = 2xt + 3y^2te^{st} = 2st^2 + 3e^{2st}te^{st} + 2st^2 + 3te^{3st}.$$

Again, we can check by observing that $G(s,t) = (st)^2 + e^{3st}$ and performing the straightforward calculations. ◀

Many other chains are possible, of course.

Example 12.32: The schema



would lead to the chain rules

$$\begin{aligned}\frac{\partial z}{\partial r} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial r}\\ \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial s}\\ \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial u}.\end{aligned}$$

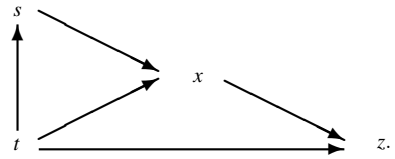
You should invent a concrete example of this schema and test out the chain rule for it. ◀

Exercises

12.5.1 Invent a concrete example to illustrate the chain rule for Example 12.32 and verify by direct computation.

12.5.2 Suppose that the dependencies of the variables can be described informally as "w depends on all three of x, y, and z and each in turn depends on both s and t." Express this schematically and write a chain rule for it.

12.5.3 Write chain rules for $\frac{\partial z}{\partial t}$ and $\frac{\partial z}{\partial s}$ that relate to the schema



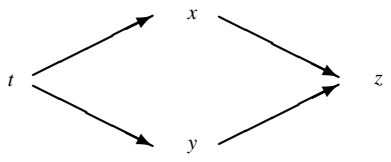
12.5.4 Let $z = F(x,y)$, $x = g(u,v,w)$, and $y = h(u,v)$.

(a) Make a path schema to show the dependencies among the variables z, x, y, u, v, and w.

(b) Write chain rules for $\frac{\partial z}{\partial u}$, $\frac{\partial z}{\partial v}$, and $\frac{\partial z}{\partial w}$.

12.5.2 Informal Proof of a Chain Rule

Let's try to see, informally, why a chain rule works. Consider, for example, the schema



and its chain rule

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

Assuming the needed differentiability requirements and using obvious notation, we can write

$$\triangle z = f_1 \triangle x + f_2 \triangle y + \varepsilon(|\triangle x| + |\triangle y|) \tag{47}$$

where $\varepsilon \to 0$ as $(\triangle x, \triangle y) \to (0,0)$. Dividing by $\triangle t$, we get

$$\frac{\triangle z}{\triangle t} = f_1 \frac{\triangle x}{\triangle t} + f_2 \frac{\triangle y}{\triangle t} + \varepsilon\left(\frac{|\triangle x|}{\triangle t} + \frac{|\triangle y|}{\triangle t}\right). \tag{48}$$

Letting $\triangle t \to 0$ in (48), we note that since $\triangle x \to 0$ and $\triangle y \to 0$ as $\triangle t \to 0$, $\varepsilon \to 0$ as $\triangle t \to 0$.

Now consider the term

$$\varepsilon\left(\frac{|\triangle x|}{\triangle t} + \frac{|\triangle y|}{\triangle t}\right)$$

as $\triangle t \to 0$. The term in parentheses approaches

$$\varepsilon\left(\left|\frac{dx}{dt}\right| + \left|\frac{dy}{dt}\right|\right) \quad \text{when } \triangle t \to 0+ \text{ and}$$

$$-\varepsilon\left(\left|\frac{dx}{dt}\right| + \left|\frac{dy}{dt}\right|\right) \quad \text{when } \triangle t \to 0- .$$

Since $\varepsilon \to 0$, the product approaches 0 as $\triangle t \to 0$ from either side. Thus

$$\frac{dz}{dt} = \lim_{\triangle t \to 0} \frac{\triangle z}{\triangle t} = f_1 \frac{dx}{dt} + f_2 \frac{dy}{dt} + 0 = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

You will note that this development presents merely the idea of a proof. We haven't spelled out various hypotheses. For example, (47) requires differentiability of z (as a function of x and y). At this point, we are trying only to set things up so we can proceed with rigorous proofs of some chain rules in the next subsection. There we shall prove a chain rule that covers Example 12.30. Proofs of other chain rules would follow similar patterns.

Exercises

12.5.5 Write up an informal proof of a different chain rule than the one here.

12.5.3 Notation of Chain Rules

A word of caution about notation is needed. Chains such as

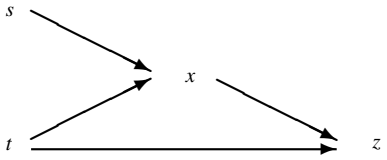
$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

are convenient to write using symbols such as $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ because the notation is familiar and suggests various "cancelations:" Thus we are tempted to say

$$\frac{\partial z}{\partial x}\frac{\partial x}{\partial s} \text{ "looks like" } \frac{\partial z}{\partial s}.$$

But care must be taken when variables appear at different levels.

Consider the schema



and its associated chain rule

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial t} + \frac{\partial z}{\partial x}\frac{\partial x}{\partial t}. \tag{49}$$

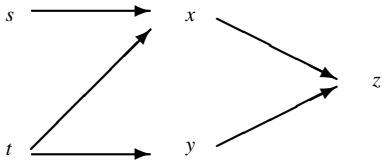
The symbol $\frac{\partial z}{\partial t}$ has two different meanings here, one on the left side of (49), the other on the right side. Let's sort this out with a concrete example.

Example 12.33: Let $z = te^{st}$. If we compute $\frac{\partial z}{\partial t}$ directly,we obtain from the product rule that $\frac{\partial z}{\partial t} = e^{st} + ste^{st}$. Now let's view z as a composite function. Let $x = e^{st}$, so $z = tx$. From the chain rule (49) we calculate

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial t} + \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} \\ &= x + t\frac{\partial x}{\partial t} = e^{st} + tse^{st}\end{aligned}$$

as before. The first occurrence of $\frac{\partial z}{\partial t}$ is what we are after. It represents the partial of z with respect to t when we view z as a function of s and t. The second occurrence represents the partial of z with respect to t when we view z as a function of x and t. ◀

If a complicated schema causes confusion, it would be preferable to use other notation that avoids ambiguity. We can avoid the ambiguous notation in (49) by introducing an additional variable $y = t$. The schema then becomes



and the chain becomes

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

where

$$\frac{\partial y}{\partial t} = \frac{dy}{dt} = 1$$

since $y = t$. In this expression $\frac{\partial z}{\partial y}$ eliminates use of $\frac{\partial z}{\partial t}$ in an ambiguous manner. Or we could use unambiguous notation by writing

$$z = F(x,t)\ ,\ x = f(s,t),$$

so $z = F(f(s,t),t)$. Then

$$\frac{\partial z}{\partial t} = F_1(f(s,t),t)f_2(s,t) + F_2(f(s,t),t).$$

Exercises

12.5.6 Suppose that $z = f(x,y)$ and $y = e^x$. Then by the chain rule

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial x} = \frac{\partial z}{\partial x} + e^x\frac{\partial z}{\partial y}.$$

A careless student suggests that we cancel $\frac{\partial z}{\partial x}$ on both sides and get $e^x\frac{\partial z}{\partial y} = 0$ so $\frac{\partial z}{\partial y} = 0$. Do you agree with this? If not, what is the correct computation of $\frac{\partial z}{\partial y}$ and what are the appropriate hypotheses you need?

12.5.7 Let F be differentiable on $\mathbb{R}^2$. Let $x = r\cos\theta$, $y = r\sin\theta$ and let

$$G(r,\theta) = F(x,y) = F(r\cos\theta, r\sin\theta).$$

Thus F is transformed into G when we transform rectangular coordinates into polar coordinates. Show that

$$\frac{\partial G}{\partial r} = \frac{\partial F}{\partial x}\cos\theta + \frac{\partial F}{\partial y}\sin\theta$$

and

$$\frac{\partial G}{\partial \theta} = -\frac{\partial F}{\partial x}r\sin\theta + \frac{\partial F}{\partial y}r\cos\theta.$$

12.5.8 Consider a rectangle with horizontal side x and vertical side y. Its area is given by the formula $A(x,y) = xy$. Its perimeter is $P(x,y) = 2x + 2y$. We readily compute

$$\frac{\partial A}{\partial x} = y. \tag{50}$$

Now let us view A as a function of x and P.

$$A(x,P) = x\left(\frac{P}{2} - x\right) = \frac{Px}{2} - x^2.$$

Here $\frac{\partial A}{\partial x} = \frac{P}{2} - 2x$, which in terms of x and y gives

$$\frac{\partial A}{\partial x} = y - x. \tag{51}$$

We see that (50) and (51) don't agree.

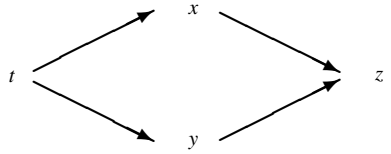
(a) Explain the apparent discrepancy.

(b) Obtain (50) by viewing A as a function of x and y via the intermediate variable P.

SEE NOTE 282

12.5.4 Proofs of Chain Rules (I)

In Section 12.5.2 we provided a nonrigorous indication why chain rules work. We considered in Example 12.28 the case corresponding to the chain



We now give a precise statement and proof of this chain rule that can serve as a model for how to proceed in general.

Theorem 12.34: *Let f and g be real-valued functions defined on a neighborhood of $t_0 \in \mathbb{R}$. Suppose both f and g are differentiable at t_0. Let F be a real-valued function defined on a neighborhood of $(x_0, y_0) \in \mathbb{R}^2$, where $x_0 = f(t_0)$ and $y_0 = g(t_0)$. Suppose F is differentiable at (x_0, y_0). Let $G(t) = F(f(t), g(t))$. Then G is differentiable at t_0 and*

$$G'(t_0) = F_1(x_0, y_0) f'(t_0) + F_2(x_0, y_0) g'(t_0).$$

Proof. Write $x = f(t)$, $y = g(t)$. Then

$$\begin{aligned} G(t) - G(t_0) &= F(f(t), g(t)) - F(f(t_0), g(t_0)) \\ &= F(x, y) - F(x_0, y_0). \end{aligned}$$

Since F is differentiable at (x_0, y_0), we can write

$$\begin{aligned} \frac{G(t) - G(t_0)}{t - t_0} &= \frac{F(x, y) - F(x_0, y_0)}{t - t_0} \\ &= F_1(x_0, y_0) \frac{(x - x_0)}{t - t_0} + F_2(x_0, y_0) \frac{(y - y_0)}{t - t_0} \\ &+ \varepsilon(x, y) \left(\frac{|x - x_0|}{t - t_0} + \frac{|y - y_0|}{t - t_0} \right) \end{aligned} \tag{52}$$

where

$$\varepsilon(x, y) \to 0 \text{ as } |x - x_0| \to 0 \text{ and } |y - y_0| \to 0. \tag{53}$$

We now let $t \to t_0$ and consider the three terms in lines 2 and 3 of (52). Since f and g are differentiable at t_0, they are continuous there, so

$$f(t) \to f(t_0) \text{ and } g(t) \to g(t_0) \text{ as } t \to t_0.$$

Thus

$$x \to x_0 \text{ and } y \to y_0 \text{ as } t \to t_0. \tag{54}$$

We also have

$$\frac{x - x_0}{t - t_0} = \frac{f(t) - f(t_0)}{t - t_0} \to f'(t_0) \text{ as } t \to t_0 \text{ and} \tag{55}$$

$$\frac{y - y_0}{t - t_0} = \frac{g(t) - g(t_0)}{t - t_0} \to g'(t_0) \text{ as } t \to t_0. \tag{56}$$

The last term of (52) is a product of two terms. The first term is $\varepsilon(x, y)$, which approaches zero as $t \to t_0$ [by (53) and (54)]. the second term is

$$\frac{|x - x_0|}{t - t_0} + \frac{|y - y_0|}{t - t_0},$$

which in absolute value is less than or equal to

$$\left| \frac{x - x_0}{t - t_0} \right| + \left| \frac{y - y_0}{t - t_0} \right|.$$

This last expression approaches $|f'(t_0)| + |g'(t_0)|$, so

$$\varepsilon(x, y) \left(\frac{|x - x_0|}{t - t_0} + \frac{|y - y_0|}{t - t_0} \right) \to 0 \text{ as } t \to t_0. \tag{57}$$

Thus, we see from (52), (55), (56), and (57) that

$$\lim_{t\to t_0}\frac{G(t)-G(t_0)}{t-t_0}=F_1(x_0,y_0)f'(t_0)+F_2(x_0,y_0)g'(t_0)$$

as was to be proved. ■

Exercises

12.5.9 Formulate and prove a version of Theorem 12.34 that would apply to a function $G(t)=F(f(t),g(t),h(t))$ and conclude that G is differentiable at t_0.

12.5.5 Mean Value Theorem

Theorem 12.34 allows us to obtain a two-dimensional analogue of the classical mean value theorem. Recall that the mean value theorem asserts, under appropriate differentiability assumptions on a function f, that

$$f(x_0+h)-f(x_0)=f'(\xi_0)h$$

for some ξ_0 between x_0+h and x_0.

Theorem 12.35: *Let F be defined on an open set $D\subset\mathbb{R}^2$. Let (x_0,y_0) and (x_0+h,y_0+k) be points in D, and suppose the line segment L determined by these points lies in D. Suppose F is differentiable on L. Then there exist ξ_0 between x_0 and x_0+h and η_0 between y_0 and y_0+k such that*

$$F(x_0+h,y_0+k)-F(x_0,y_0)=F_1(\xi_0,\eta_0)h+F_2(\xi_0,\eta_0)k. \tag{58}$$

Proof. We begin by expressing the line segment L parametrically:

$$x=x_0+th,\ y=y_0+tk\quad(0\le t\le 1).$$

We thus can view the function F on L as a function of t on $[0,1]$. Let

$$G(t)=F(x_0+th,y_0+tk),\quad(0\le t\le 1). \tag{59}$$

By Theorem 12.34

$$G'(t)=hF_1(x_0+th,y_0+tk)+kF_2(x_0+th,y_0+tk). \tag{60}$$

By the mean value theorem (Theorem 7.20) for functions of one variable we obtain

$$G(1)-G(0)=G'(t_0)\ \text{ for some }\ t_0\in(0,1).$$

Now, we see from (59) that $G(0)=F(x_0,y_0)$,

$$G(1)=F(x_0+h,y_0+k),$$

so

$$G'(t_0)=F(x_0+h,y_0+k)-F(x_0,y_0). \tag{61}$$

But from (60) we have

$$G'(t_0)=hF_1(x_0+t_0h,y_0+t_0k)+kF_2(x_0+t_0h,y_0+t_0k). \tag{62}$$

Combining (61) and (62) we get

$$F(x_0+h,y_0+k)-F(x_0,y_0)= \\ hF_1(x_0+t_0h,y_0+t_0k)+kF_2(x_0+t_0h,y_0+t_0k).$$

To obtain (58), we simply let $\xi_0 = x_0+t_0h$ and $\eta_0 = y_0+t_0k$. ■

Exercises

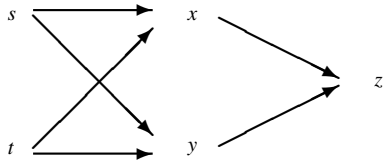
12.5.10 State the mean value theorem (Theorem 12.35) in a form that gives the conclusion

$$F(P_2)-F(P_1) = \bigtriangledown F(P_0)\cdot(P_1-P_2)$$

where P_1, P_2, and P_0 are points in $\mathbb{R}^2$ and $\bigtriangledown$ is the gradient.

12.5.6 Proofs of Chain Rules (II)

We now turn to a statement and proof of a chain rule that corresponds to the schema in Example 12.30.



Theorem 12.36: *Let F be defined on an open set $D \subset \mathbb{R}^2$ and let (x_0,y_0) be a point in D. Suppose F is differentiable at (x_0,y_0). Let f and g be defined in a neighborhood of the point $(s_0,t_0) \in \mathbb{R}^2$. Suppose f and g have first partial derivatives at (s_0,t_0) and that $x_0 = f(s_0,t_0)$, $y_0 = g(s_0,t_0)$. Define G by $G(s,t) = F(f(s,t),g(s,t))$. Then G has first partial derivatives at (s_0,t_0) and*

$$\left.\begin{array}{rcl} G_1(s_0,t_0) & = & F_1(x_0,y_0)f_1(s_0,t_0)+F_2(x_0,y_0)g_1(s_0,t_0) \\ G_2(s_0,t_0) & = & F_1(x_0,y_0)f_2(s_0,t_0)+F_2(x_0,y_0)g_2(s_0,t_0) \end{array}\right\}. \tag{63}$$

Before proving Theorem 12.36 we make several observations.

1. You should check that formulas (63) are just precise versions of the schema in Example 12.30. In particular, formulas (63) make it clear where the partials are evaluated.

2. In Theorem 12.34 we concluded differentiability of the function G. Here, we don't. Nor do we assume differentiability of f and g here. Theorem 12.36 is simply a theorem about first partial derivatives of G, their existence, and a formula for them. [It is true that if we assumed differentiability of f and g at (s_0,t_0), then we could have concluded that G is differentiable, but we shall not prove this until Section 12.8.]

3. Note the similarity between the approaches to the proofs of Theorems 12.34 and 12.36.

Proof. We shall establish the first of the rules (63), the proof of the second being similar (Exercise 12.5.11).

To compute $G_1(s_0,t_0)$, write $x = f(s,t_0)$, $y = g(s,t_0)$. Then by definition of G,

$$G(s,t_0) - G(s_0,t_0) = F(x,y) - F(x_0,y_0). \tag{64}$$

Since F is assumed differentiable at (x_0,y_0) we can write

$$\begin{aligned} F(x,y) - F(x_0,y_0) = & \\ & F_1(x_0,y_0)(x-x_0) + F_2(x_0,y_0)(y-y_0) \\ & +\varepsilon(x,y)(|x-x_0| + |y-y_0|) \end{aligned} \tag{65}$$

where $\varepsilon(x,y) \to 0$ as $(x,y) \to (x_0,y_0)$.

From (64) and (65) we see that, for $s \neq s_0$,

$$\begin{aligned} \frac{G(s,t_0) - G(s_0,t_0)}{s-s_0} = \frac{F(x,y) - F(x_0,y_0)}{s-s_0} = & \\ F_1(x_0,y_0)\frac{x-x_0}{s-s_0} + F_2(x_0,y_0)\frac{y-y_0}{s-s_0} + & \\ \varepsilon(x,y)\left(\frac{|x-x_0|}{s-s_0} + \frac{|y-y_0|}{s-s_0}\right). & \end{aligned} \tag{66}$$

We can now complete the proof (as we did in the proof of Theorem 12.34) by letting $s \to s_0$. The right side of (66) approaches

$$F_1(x_0,y_0)f_1(s_0,t_0) + F_2(x_0,y_0)g_1(s_0,t_0), \tag{67}$$

the fact that the remaining term

$$\varepsilon(x,y)\left(\frac{|x-x_0|}{s-s_0} + \frac{|y-y_0|}{s-s_0}\right)$$

approaches zero being similar to the corresponding part of the proof of Theorem 12.34. As a result, the left side of (66) approaches a limit as $s \to s_0$. That limit is, of course, $G_1(s_0,t_0)$. Thus

$$G_1 = F_1(x_0,y_0)f_1(s_0,t_0) + F_2(x_0,y_0)g_1(s_0,t_0).$$

■

Exercises

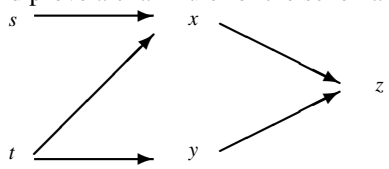
12.5.11 Prove the second of the chain rules (63).

12.5.12 Provide the details of the argument that the term

$$\varepsilon(x,y)\left(\frac{|x-x_0|}{s-s_0} + \frac{|y-y_0|}{s-s_0}\right)$$

in (66) approaches zero as $s \to s_0$.

12.5.13 State precisely and prove a chain rule for the schema



in two ways: First, by imitating the proofs of Theorems 12.34 and 12.36; then, by viewing this schema as a special case of the schema governing Theorem 12.36.

12.5.7 Higher Derivatives

It is sometimes necessary to use a chain rule to calculate higher partial derivatives. Consider, for example, Laplace's equation

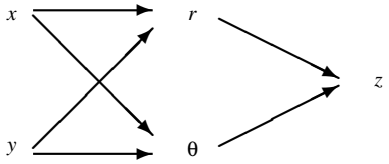
$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0. \tag{68}$$

A twice differentiable function z satisfying this equation is called *harmonic*. Such functions are important in many parts of mathematics (such as complex analysis, applied mathematics) and physics. The expression

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}$$

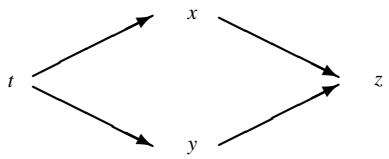
is called the *Laplacian* of the function z.

Suppose we wish to express this equation in polar coordinates. We let $x = r\cos\theta$, $y = r\sin\theta$, and consider the following schema.



We want to express (68) in terms of r and θ by obtaining $\frac{\partial^2 z}{\partial x^2}$ and $\frac{\partial^2 z}{\partial y^2}$ as functions of r and θ. The calculations are messy and have been left as Exercise 12.5.14.

We shall instead consider a less messy schema first shown in Example 12.28.



(69)

Here $z = F(x,y)$, while x and y are functions of t. The corresponding chain rule is

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}. \tag{70}$$

We wish to compute $\frac{d^2 z}{dt^2} = \frac{d}{dt}\left(\frac{dz}{dt}\right)$. From (70) we see that

$$\begin{aligned} \frac{\partial^2 z}{\partial t^2} &= \frac{d}{dt}\left(\frac{\partial z}{\partial x}\frac{dx}{dt}\right) + \frac{d}{dt}\left(\frac{\partial z}{\partial y}\frac{dy}{dt}\right) \\ &= \frac{d}{dt}\left(\frac{\partial z}{\partial x}\right)\frac{dx}{dt} + \frac{\partial z}{\partial x}\frac{d^2 x}{dt^2} \\ &+ \frac{d}{dt}\left(\frac{\partial z}{\partial y}\right)\frac{dy}{dt} + \frac{\partial z}{\partial y}\frac{d^2 y}{dt^2}. \end{aligned} \tag{71}$$

This last expression involves two terms that should be further developed, $\frac{d}{dt}\left(\frac{\partial z}{\partial x}\right)$, and $\frac{d}{dt}\left(\frac{\partial z}{\partial y}\right)$. Now $\frac{\partial z}{\partial x} = F_1$ and $\frac{\partial z}{\partial y} = F_2$. Both of these are functions of x and y. To obtain their derivatives with respect to t, we note that the schema (69) applies again. We obtain

$$\begin{aligned}\frac{d}{dt}\left(\frac{\partial z}{\partial x}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)\frac{dx}{dt} + \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)\frac{dy}{dt} \\ &= \frac{\partial^2 z}{\partial x^2}\frac{dx}{dt} + \frac{\partial^2 z}{\partial y \partial x}\frac{dy}{dt}.\end{aligned} \tag{72}$$

Similarly,

$$\frac{d}{dt}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2}\frac{dy}{dt} + \frac{\partial^2 z}{\partial x \partial y}\frac{dx}{dt}. \tag{73}$$

If we assume continuity of all partials involved, the mixed partials appearing in (72) and (73) are equal (Theorem 12.5). Substituting (72) and (73) into (71) and rearranging the terms, leads to the formula

$$\begin{aligned}\frac{\partial^2 z}{\partial t^2} = \frac{\partial^2 z}{\partial x^2}\left(\frac{dx}{dt}\right)^2 + 2\frac{\partial^2 z}{\partial x \partial y}\frac{dx}{dt}\frac{dy}{dt} \\ + \frac{\partial^2 z}{\partial y^2}\left(\frac{dy}{dt}\right)^2 + \frac{\partial z}{\partial x}\frac{d^2 x}{dt^2} + \frac{\partial z}{\partial y}\frac{d^2 y}{dt^2}.\end{aligned} \tag{74}$$

If we do not assume sufficient regularity of the partials to assure the mixed partials are equal, the second term of (74) must be expressed as the sum

$$\left(\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y \partial x}\right)\frac{dx}{dt}\frac{dy}{dt}.$$

Example 12.37: Formula (74) can be illustrated with the simple example

$$z = x^2 y^3 \ , \ x = t^4 \ , \ y = t^5.$$

Thus directly we obtain $z = t^{23}$, so $\frac{d^2 z}{dt^2} = 506t^{21}$. Application of formula (74) gives the same result. ◀

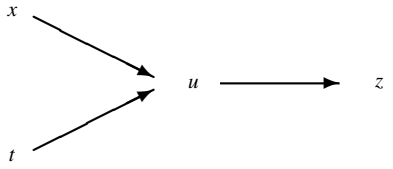
Example 12.38: Consider the wave equation

$$c^2\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial t^2},$$

where c is a constant. Here $f(x,t)$ describes the vertical displacement of a particle in a wave corresponding to the horizontal coordinate x at time t. We show that any function of the form

$$f(x,t) = g(x+ct),$$

where g is twice differentiable, satisfies this equation. By making the substitutions $u = x+ct$, $z = g(u)$ we arrive at the following schema, which is equivalent to schema (43) of Example 12.26.



Thus the appropriate chain rule becomes

$$f_1(x,t) = g'(u)\frac{\partial u}{\partial x} = g'(x+ct)$$
$$f_2(x,t) = g'(u)\frac{\partial u}{\partial t} = cg'(x+ct)$$

Applying the chain rule again, we obtain

$$f_{11}(x,t) = \frac{\partial}{\partial x}(g'(x+ct)) = g''(x+ct),$$
$$f_{22}(x,t) = \frac{\partial}{\partial t}(cg'(x+ct)) = c^2 g''(x+ct).$$

Thus $c^2 f_{11}(x,t) = f_{22}(x,t)$, or

$$c^2\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial t^2},$$

verifying that the function f does satisfy the wave equation. ◀

Exercises

12.5.14 Verify that the Laplacian (68) transformed into polar coordinates via the substitution $x = r\cos\theta$, $y = r\sin\theta$ becomes

$$\frac{\partial^2 z}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 z}{\partial \theta^2} + \frac{1}{r}\frac{\partial z}{\partial r}.$$

12.5.15 The Laplacian for functions of three variables takes the form

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial^2 z}.$$

Transform this expression to one in cylindrical coordinates via the substitutions $x = r\cos\theta$, $y = r\sin\theta$, and $z = z$.

SEE NOTE 283

12.5.16 Use the result of Exercise 12.5.15 to obtain the Laplacian in spherical coordinates.

SEE NOTE 284

12.6 Implicit Function Theorems

Suppose that we are required to solve an equation of the form

$$F(x,y) = 0 \tag{75}$$

for y as a function of x. What we must mean by stating that some function $y = \phi(x)$ "solves" this equation for y in terms of x in a neighborhood I of a point x_0 is that $\phi : I \to \mathbb{R}$ and $F(x,\phi(x)) = 0$ for all $x \in I$. Since it is not often possible to solve such an equation explicitly [i.e., to find a formula for $\phi(x)$], we need to know conditions under which a solution does exist, conditions under which the solution is unique, and we need to know methods for obtaining the derivative $y' = \phi'(x)$ of the solution.

This should be a familiar problem. It is taught in most calculus courses as "implicit differentiation," where techniques for finding the derivative are obtained. Usually little

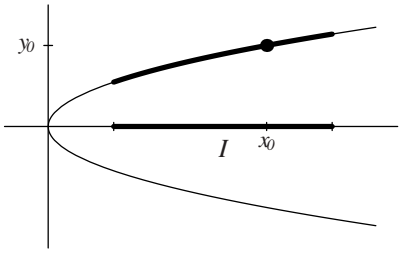


Figure 12.6. A solution of $F(x,y) = y^2 - x = 0$ in a neighborhood I of x_0.

attention is paid to the existence and uniqueness problems. An example should be enough to recall the ideas.

Example 12.39: Consider the simple equation

$$F(x,y) = y^2 - x = 0. \tag{76}$$

If we solve this equation explicitly for x in terms of y, we obtain

$$x = y^2. \tag{77}$$

In elementary calculus it is said that equation (76) presents x as a function of y *implicitly*, and that (77) gives an *explicit* representation of x as a function of y.

If, instead, we attempt to solve (76) for y in terms of x, we obtain

$$y = \pm\sqrt{x}. \tag{78}$$

This does *not* present y as a function of x because for every value of $x > 0$ there exist *two* possible values of y. Nonetheless, if $x_0 > 0$ and y_0 is one of the values that allows (76) to be satisfied, there is a neighborhood I of x_0 in $\mathbb{R}$ such that one of the choices in equation (78) represents y as a continuous function of x in that neighborhood (Fig. 12.6). Observe that we cannot make the same statement for $x_0 = 0$. Observe also that at $(0,0)$ there is a vertical tangent and, for $F(x,y) = y^2 - x$, $F_2(0,0) = 0$. ◀

The implicit function theorem that we now proceed to state and prove exactly describes this situation and gives a condition under which a solution does exist. It also justifies the calculus technique of implicit differentiation that is used to obtain the derivative of the function defined implicitly.

12.6.1 One-Variable Case

Theorem 12.40 provides a condition under which an equation of the form $F(x,y) = 0$ can locally be solved uniquely. In addition, it shows that the regularity conditions we impose on F guarantee that the solution function will also be well behaved. We view this theorem as a "warm-'up" for the more general implicit function theorems we obtain in the next two subsections. Note the *local* character of the conclusion. We do not claim a global solution.

Theorem 12.40: *Let D be an open set in $\mathbb{R}^2$ and let $F : D \to \mathbb{R}$. Suppose F has continuous partial derivatives F_1 and F_2 on D. Let $(x_0, y_0) \in D$ be such that*

$$F(x_0, y_0) = 0 \text{ and } F_2(x_0, y_0) \neq 0.$$

Then there is an open interval $I_0 \in \mathbb{R}$ and a continuously differentiable function $\phi : I_0 \to \mathbb{R}$ such that $x_0 \in I_0$, $(x, \phi(x)) \in D$ for all $x \in I_0$, $\phi(x_0) = y_0$, and such that $F(x, \phi(x)) = 0$ for all $x \in I_0$. Furthermore, the formula

$$\phi'(x) = -\frac{F_1(x, \phi(x))}{F_2(x, \phi(x))}$$

is valid for all $x \in I_0$.

Proof. Suppose $F_2(x_0, y_0) > 0$. [The proof is similar if $F_2(x_0, y_0) < 0$.] Since F_2 is continuous by assumption, there exists a neighborhood $N \subset D$ of (x_0, y_0) such that $F_2 > 0$ on N. We may take N to be rectangular of the form $N = I \times J$, with (x_0, y_0) the center of the rectangle. Suppose $J = [c, d]$. Since $F_2 > 0$ on N, the function $F(x_0, \cdot)$ is increasing on J. Since $F(x_0, y_0) = 0$, it follows that

$$F(x_0, c) < 0 < F(x_0, d).$$

The function F is continuous, so there exists an open interval $I_0 \subset I$ such that x_0 is the center of I_0 and

$$F(x, c) < 0 < F(x, d) \text{ for each } x \in I_0. \tag{79}$$

The continuity of F (as a function of two variables) implies that for each $x \in I_0$ the function $F(x, \cdot)$ is continuous in the second variable. It follows from (79) and the intermediate value property of continuous functions (which we discussed in Section 5.8) that for each $x \in I_0$ there exists at least one value $y \in (c, d)$ such that $F(x, y) = 0$. Moreover, since $F(x, \cdot)$ is also increasing, this value of y is unique. We can express this dependency of y on x by writing $y = \phi(x)$. Thus we have found a function ϕ such that for each $x \in I_0$,

$$F(x, \phi(x)) = 0 \tag{80}$$

and

$$c < \phi(x) < d. \tag{81}$$

We now show that ϕ has all the remaining properties claimed in the conclusion of the theorem. It is clear that $\phi(x_0) = y_0$ and that $(x, \phi(x)) \in D$ for each $x \in I_0$.

We next check that ϕ is continuous on I_0. Observe first from (81) that if $x_1, x_2 \in I_0$, then $|\phi(x_1) - \phi(x_2)| < d - c$. The rectangle N could have been chosen as small as we like, a smaller rectangle leading perhaps to a shorter interval I_0. Thus, if x_1 is any point in I_0 and $\varepsilon > 0$, we can apply the same argument we gave previously on a neighborhood $I_1 \subset I_0$ of x_1 with $d_1 - c_1 < \varepsilon$. In Figure 12.7 we have added the rectangle $I_0 \times J$ centered at (x_0, y_0) to Figure 12.6. The result is that $|\phi(x) - \phi(x_1)| < \varepsilon$ on I_1. But this implies ϕ is continuous at x_1. Since x_1 is an arbitrary point of I_1, ϕ is continuous on I_0.

It remains to show that ϕ is continuously differentiable on I_0 and that the formula given for ϕ' is valid. Let $x, x + h \in I_0$. Since the graph of ϕ is contained in the rectangle

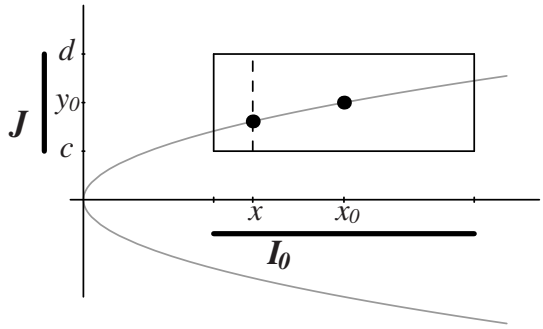


Figure 12.7. Construction of the rectangle $I_0 \times J$ in the proof.

N, the line segment L with endpoints $(x,\phi(x))$ and $(x+h,\phi(x+h))$ lies in N. By the mean value theorem (Theorem 12.35) there exists $(\xi,\eta) \in L$ such that

$$F(x+h,\phi(x+h)) - F(x,\phi(x)) = \\ F_1(\xi,\eta)h + F_2(\xi,\eta)[\phi(x+h) - \phi(x)].$$

But $F(x+h,\phi(x+h)) = F(x,\phi(x)) = 0$, so

$$F_1(\xi,\eta)h + F_2(\xi,\eta)[\phi(x+h) - \phi(x)] = 0.$$

Thus, if $F_2(\xi,\eta) \neq 0$, we can write

$$\frac{\phi(x+h)-\phi(x)}{h} = -\frac{F_1(\xi,\eta)}{F_2(\xi,\eta)}. \tag{82}$$

Since F_2 is never 0 in some neighborhood of $(x,\phi(x))$, equation (82) is valid in that neighborhood. As $h \to 0$, the left side of (82) approaches $\phi'(x)$, (ξ,η) approaches $(x,\phi(x))$, and, because of the continuity of F_1 and F_2, the right side of (82) approaches

$$-\frac{F_1(x,\phi(x))}{F_2(x,\phi(x))},$$

as was to be proved. ■

Example 12.41: The example $F(x,y) = x^2 - y^4$ and $(x_0,y_0) = (1,1)$ illustrates that while there is a solution valid in a neighborhood of $x_0 = 1$ there cannot exist a solution in too large an interval. Indeed there is no solution in any interval I_0 that contains $x = 0$. ◄

Example 12.42: We return to Example 12.39:

$$F(x,y) = y^2 - x = 0.$$

Here $F_1(x,y) = -1$, $F_2(x,y) = 2y$. The hypotheses of Theorem 12.40 are met provided that $y \neq 0$. If $(x_0,y_0) = (4,2)$, then the resulting function is $\phi(x) = \sqrt{x}$ and $\phi'(x) = 1/2y$. In this case, the interval I_0 can be any interval containing $x_0 = 4$ but not containing 0. ◄

Exercises

12.6.1 Show that the equation $x^2y^2 + 2e^{xy} - 4 - 2e^2 = 0$ can be solved for y in terms of x in a neighborhood of the point $x = 1$ with $y(1) = 2$. Calculate $\frac{dy}{dx}$ when $x = 1$.

12.6.2 Do Exercise 12.6.1 via implicit differentiation as one would in an elementary calculus class. (There one usually simply *assumes* the existence of a solution.)

12.6.2 Several-Variable Case

Theorem 12.40 extends to situations involving more variables. Consider one equation in three variables. We wish to solve an equation

$$F(x,y,z) = 0$$

for z as a function of x and y in some neighborhood of a point (x_0,y_0). As before, the solution should be a function $z = \phi(x,y)$ so that

$$F(x,y,\phi(x,y)) = 0$$

for all (x,y) in that neighborhood.

Example 12.43: Consider the equation

$$x^2 + y^2 + z^2 - 9 = 0. \tag{83}$$

Equation (83) can easily be untangled to represent one of the variables in terms of the others. Thus

$$z = \pm\sqrt{9 - x^2 - y^2} \tag{84}$$

represents z in terms of x and y when $x^2 + y^2 \leq 3$. Just as we saw in the one-variable case, z is not a function of x and y on the disk $x^2 + y^2 \leq 9$ because the values of z are not uniquely determined by equation (84). Nonetheless, if (x_0,y_0) is a point inside this disk and and z_0 is one of the values that allows (83) to be satisfied, there is a neighborhood I of (x_0,y_0) in $\mathbb{R}^2$ such that one of the choices in equation (84) represents z as a continuous function of (x,y) in that neighborhood. ◀

Theorem 12.44 that follows is the analogue of Theorem 12.40 when we deal with one equation in more than two variables. We state the theorem for $n+1$ variables.

Theorem 12.44: *Let F have continuous first-order partial derivatives on an open set $D \subset \mathbb{R}^{n+1}$. Let $x_0 = (x_1^0, x_2^0, \ldots, x_n^0, z_0) \in D$. Suppose*

$$F(x_1^0, x_2^0, \ldots, x_n^0, z_0) = 0$$

and

$$F_{n+1}(x_1^0, x_2^0, \ldots, x_n^0, z_0) \neq 0.$$

Then there is a neighborhood $J_0 \subset \mathbb{R}$ of z_0 and a neighborhood $I_0 \subset \mathbb{R}^n$ of $(x_1^0, x_2^0, \ldots, x_n^0)$, and there is a unique function ϕ defined on I_0 such that

$$z_0 = \phi(x_1^0, x_2^0, \ldots, x_n^0)$$

and

$$F(x_1, \ldots, x_n, \phi(x_1, \ldots, x_n)) = 0$$

for all $(x_1,\ldots,x_n) \in I_0$. *Furthermore, the function* ϕ *has continuous partial derivatives with respect to each of the variables* $(x_1,\ldots,x_n)$ *and*

$$\phi_i(x_1,\ldots,x_n) = -\frac{F_i(x_1,\ldots,x_n,z)}{F_{n+1}(x_1,\ldots,x_n,z)} \tag{85}$$

for each $i = 1,\ldots,n$ *and for all* $(x_1,\ldots,x_n) \subset I_0$.

Proof. Observe that by letting $z = \phi(x_1,\ldots,x_n)$, we have explicitly represented the variable z as a continuously differentiable function of $x_1,\ldots,x_n$ on a neighborhood of $(x_1^0,\ldots,x_n^0)$ once we specify the value of z_0.

We prove Theorem 12.44 for the case $n = 2$. The general proof needs no fresh ideas, but the notation is more messy and pictures are less easy to visualize. Observe the similarity of the ideas underlying our proof and the proof of Theorem 12.40. To simplify notation, let us denote the point in question by (x_0,y_0,z_0). We first obtain the neighborhoods J_0 and I_0 mentioned in the theorem.

Assume that $F_3(x_0,y_0,z_0) > 0$. [The case $F_3(x_0,y_0,z_0) < 0$ is similar.] Since F_3 is continuous in D, there is an open rectangular box, A, with center (x_0,y_0,z_0) and edges parallel to the coordinate axes such that $F_3 > 0$ on A. Let $2c$ be the height of this box. Then $F(x_0,y_0,\cdot)$ is an increasing function of z on the interval $[z_0-c,z_0+c]$. Since $F(x_0,y_0,z_0) = 0$ by assumption, we have

$$F(x_0,y_0,z_0-c) < 0 < F(x_0,y_0,z_0+c).$$

Now F is continuous on D since, by assumption, F has continuous first partial derivatives on D. We can thus obtain positive numbers a and b such that

$$F_1(x,y,z_0-c) < 0 \text{ and } F(x,y,z_0+c) > 0$$

if $|x-x_0| < a$ and $|y-y_0| < b$. We can choose a and b so small that the box

$$B = \{(x,y,z) : |x-x_0| < a, |y-y_0| < b, |z-z_0| < c\}$$

is contained in A. The required neighborhoods are thus

$$I_0 = \{(x,y) : |x-x_0| < a, |y-y_0| < b\}$$

and

$$J_0 = \{z : |z-z_0| < c\}.$$

We next obtain the function ϕ. Let $(x,y) \in I_0$. Since $F_3 > 0$ on the set $B = I_0 \times J_0$, it follows that $F(x,y,z)$ is a continuous increasing function of z on the interval $[z_0-c,z_0+c]$. It is positive at (x,y,z_0+c) and negative at (x,y,z_0-c), so there is a unique value of $z \in [z_0-c,z_0+c]$ such that $F(x,y,z) = 0$. Denote this value of z by $\phi(x,y)$. We have thus determined the function ϕ on I_0.

We now show that ϕ has the desired properties. We prove continuity as we did in Theorem 12.40 . Let $(x_1,y_1) \in I_0$ and let $\varepsilon > 0$. We may assume $\varepsilon \le c$. Let $z_1 = \phi(x_1,y_1)$. Then $F(x_1,y_1,z_1) = 0$. Applying the same argument that we applied earlier to obtain the box B, we obtain a (possibly smaller) box B_1, centered at (x_1,y_1,z_1), whose height, $2c_1$, is as small as we like. Take $2c_1 < \varepsilon$ and let $B_1 = I_1 \times J_1$ were

$$J_1 = [z_1-c_1,z_1+c_1]$$

and

$$I_1 = \{(x,y) : |x-x_1| < a_1, |y-y_1| < b_1\}.$$

Our previous argument applied to B_1 gives rise to a function g defined on I_1 such that $F(x,y,g(x,y)) = 0$ on I_1 and

$$z_1 - c_1 < g(x,y) < z_1 + c_1$$

on I_1. Thus, if $(\xi,\eta) \in I_1$, then

$$|g(\xi,\eta) - g(x_1,y_1)| < 2c_1 < \varepsilon.$$

It follows that g is continuous at (x_1,y_1). But g was obtained from F in exactly the same way ϕ was, and the values are unique, so $g = \phi$ on I_1. Thus ϕ is continuous at (x_1,y_1). Since (x_1,y_1) is an arbitrary point of I_0, ϕ is continuous on I_0.

Finally, we verify that the formulas (85) for the partial derivatives are valid. We provide a proof for ϕ_1. The proof for ϕ_2 is the same.

Consider the function $F(\cdot,y,\cdot)$ as a function of x and z with y fixed. Write $z+k = \phi(x+h,y)$. [Here we assume, of course, that $(x,y) \in I_0$, and $(x+h,y) \in I_0$.] We may also assume $z+k \in J_0$, since ϕ is continuous on I_0. Now

$$F(x,y,z) = 0 = F(x+h,y,z+k).$$

By the mean value theorem (Theorem 12.35)

$$0 = F(x+h,y,z+k) - F(x,y,z) = F_1(\xi,y,\zeta)h + F_3(\xi,y,\zeta)k, \tag{86}$$

where ξ is between x and $x+h$ and ζ is between z and $z+k$. Thus

$$\begin{aligned} \phi_1(x,y) &= \lim_{h\to 0} \frac{\phi(x+h,y) - \phi(x,y)}{h} \\ &= \lim_{h\to 0} \frac{(z+k)-z}{h} = \lim_{h\to 0} \frac{k}{h}. \end{aligned}$$

By (86) this last limit is just

$$\lim_{h\to 0} \left(-\frac{F_1(\xi,y,\zeta)}{F_3(\xi,y,\zeta)} \right) = \frac{F_1(x,y,z)}{F_3(x,y,z)}$$

since $(\xi,\zeta) \to (x,z)$ as $h \to 0$ and the functions F_1 and F_3 are continuous by assumption. ■

Example 12.45: Returning to Example 12.43, where

$$x^2 + y^2 + z^2 - 9 = 0,$$

we find that

$$\begin{aligned} F_1(x,y,z) &= 2x, \\ F_2(x,y,z) &= 2y, \\ F_3(x,y,z) &= 2z. \end{aligned}$$

When $z \neq 0$ and $x^2+y^2+z^2 \leq 9$ we have $F_3(x,y,z) \neq 0$ and the hypotheses of theorem 12.44 are met. Given (x_0,y_0,z_0) such that $x_0^2+y_0^2+z_0^2 \leq 9$ and $z_0 \neq 0$, there exists a neighborhood N of (x_0,y_0) and a function ϕ defined on that neighborhood such that

$$F(x,y,\phi(x,y)) = 0$$

or, equivalently,

$$x^2 + y^2 + [\phi(x,y)]^2 - 9 = 0.$$

The partial derivatives of the function ϕ are given, for all $(x,y) \in N$, by

$$\phi_1(x,y) = -x/z \text{ and } \phi_2(x,y) = -y/z.$$

In fact we know that either

$$\phi(x,y) = \sqrt{9-x^2-y^2} \text{ or } \phi(x,y) = -\sqrt{9-x^2-y^2}$$

depending on whether $z_0 > 0$ or $z_0 < 0$. ◀

Exercises

12.6.3 Compare the proof of Theorem 12.44 of this section with that of Theorem 12.40 of the previous section. Were any nonobvious new ideas needed to prove Theorem 12.44?

12.6.4 Prove Theorem 12.44 as stated, that is, for general $n \geq 2$.

12.6.5 Suppose that $F(x,y,z) = 0$ defines x as a function of y, z and also y as a function of x, z and also z as a function of x, y. Show that (under appropriate hypotheses on F)

$$\frac{\partial z}{\partial x}\frac{\partial z}{\partial y}\frac{\partial x}{\partial z} = -1.$$

12.6.6 Look ahead to Exercise 13.11.3 for a proof of an implicit function theorem using metric space methods.

12.6.3 Simultaneous Equations

In the preceding section we dealt with one equation in several variables, say $F(x,y,z) = 0$. We found conditions under which we could solve for z in terms of x and y in a neighborhood of a point (x_0,y_0), obtaining a function $z = \phi(x,y)$ continuously differentiable on that neighborhood. We turn now to a situation that occurs frequently involving *several* simultaneous equations, say m equations in $m+n$ variables.

Example 12.46: Here are two equations

$$\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \end{aligned}$$

involving the four variables x, y, r, and θ. These equations can be viewed as ones giving a change of coordinate systems from polar coordinates to rectangular coordinates. Alternatively, we can view them as presenting x and y explicitly in terms of r and θ, or r and θ implicitly in terms of x and y. A third perspective is to view them as defining a mapping of $\mathbb{R}^2$ onto $\mathbb{R}^2$: $F(r,\theta) = (r\cos\theta, r\sin\theta)$.

In each of these perspectives there is reason to wish to express r and θ as functions of x and y. Our first interpretation would provide the equations to transform from rectangular to polar coordinates. The second interpretation merely would provide an *explicit* representation of r and θ in terms of x and y. The third interpretation would provide an inverse function to F: A function G such that $G(x,y) = (r,\theta)$.

You may recall that the equations

$$\begin{aligned} r &= \sqrt{x^2+y^2} \\ \theta &= \arctan\frac{y}{x} \end{aligned}$$

do the job when x and y are not both zero. ◀

Example 12.47: Consider the system

$$\left.\begin{aligned} A_1u+B_1v+C_1 &= 0 \\ A_2u+B_2v+C_2 &= 0 \end{aligned}\right\}, \tag{87}$$

where the A's, B's, and C's are functions of x and y. Here we have two equations in the four variables x, y, u, v.

Using ideas from elementary algebra, we can "solve" these equations for u and v in terms of x and y provided that the determinant

$$|J| = \begin{vmatrix} A_1 & B_1 \\ A_2 & B_2 \end{vmatrix} = A_1B_2 - A_2B_1 \neq 0. \tag{88}$$

The equations (87) take the form

$$\begin{aligned} F(x,y,u,v) &= 0 \\ G(x,y,u,v) &= 0. \end{aligned} \tag{89}$$

Note that

$$\begin{aligned} \frac{\partial F}{\partial u} &= A_1, \quad \frac{\partial F}{\partial v} = B_1, \\ \frac{\partial G}{\partial u} &= A_2, \quad \frac{\partial G}{\partial v} = B_2. \end{aligned}$$

The condition (88) takes the form

$$|J| = \begin{vmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{vmatrix} \neq 0. \tag{90}$$

◀

The determinant in (90) is called the *Jacobian determinant* or the determinant of the *Jacobian*

$$J = \begin{pmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{pmatrix}.$$

It is only one of many Jacobians that arise and play important roles in this section and in other parts of mathematics. We often write the Jacobian determinant in the form

$$\frac{\partial(F,G)}{\partial(u,v)}.$$

We are concerned with determining when solutions of the type Examples 12.46 and 12.47 suggest can be found, and in determining the partial derivatives of the functions obtained (for example, $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$, $\frac{\partial v}{\partial y}$ in Example 12.47).

We treat the case of a pair of simultaneous equations in four variables, as it is representative of the general situation.

Theorem 12.48: *Let F and G have continuous first partial derivatives on an open set $D \subset \mathbb{R}^4$ and let $p_0 = (x_0, y_0, u_0, v_0) \in D$ with*

$$\left.\begin{array}{rcl} F(x_0,y_0,u_0,v_0) & = & 0 \\ G(x_0,y_0,u_0,v_0) & = & 0 \end{array}\right\}. \tag{91}$$

Let

$$|J| = \frac{\partial(F,G)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{vmatrix}.$$

Suppose $|J|$ is not zero at p_0. Then there are neighborhoods I_0 and J_0 of (x_0, y_0) and (u_0, v_0), respectively, such that

(i) *To each $(x,y) \in I_0$ there corresponds a unique $(u,v) \in J_0$ such that equations (91) are satisfied at (x,y,u,v). This correspondence defines u and v as functions on I_0 by*

$$u = \phi(x,y)\ ,\ v = \psi(x,y).$$

(ii) *The functions ϕ and ψ have continuous partial derivatives on I_0 given by the formulas*

$$\frac{\partial\phi}{\partial x} = -\frac{1}{|J|}\frac{\partial(F,G)}{\partial(x,v)}, \quad \frac{\partial\psi}{\partial x} = -\frac{1}{|J|}\frac{\partial(F,G)}{\partial(u,x)},$$
$$\frac{\partial\phi}{\partial y} = -\frac{1}{|J|}\frac{\partial(F,G)}{\partial(y,v)}, \quad \frac{\partial\psi}{\partial y} = -\frac{1}{|J|}\frac{\partial(F,G)}{\partial(u,y)}.$$

Proof. We shall apply Theorem 12.44 twice, first to obtain v as a function of x, y, and u, and then to obtain u as a function of x and y. Since $|J| \neq 0$ at p_0, at least one of the partials $\frac{\partial F}{\partial v}$, $\frac{\partial G}{\partial v}$ must be different from 0 at p_0. We may assume $F_4 = \frac{\partial F}{\partial v} \neq 0$ at p_0, the argument being similar if $\frac{\partial G}{\partial v} \neq 0$ at p_0.

Now we apply Theorem 12.44 to the equation $F(x,y,u,v) = 0$. We obtain a function $v = g(x,y,u)$, defined in a neighborhood of (x_0, y_0, u_0) such that $F(x,y,u,g(x,y,u)) = 0$ on that neighborhood.

We next consider the function G. Let

$$H(x,y,u) = G(x,y,u,g(x,y,u)).$$

We have thus replaced the function G of four variables with the function H of three variables. Our task is to solve this equation for u in terms of x and y.

To do this, we use Theorem 12.44 once more. In order to check that Theorem 12.44 applies, we must show $H_3 \neq 0$ at (x_0, y_0, u_0). Applying the chain rule, we obtain

$$\begin{aligned} H_3 &= G_3 + G_4 g_3 = G_3 - G_4\frac{F_3}{F_4} \\ &= \frac{G_3F_4 - G_4F_3}{F_4} = -\frac{|J|}{F_4}. \end{aligned} \tag{92}$$

Now $|J|$ and F_4 are different from 0 in some neighborhood of p_0. Thus H_3 is finite and nonzero in that neighborhood. Applying Theorem 12.44 to the equation $H(x,y,u) = 0$, we obtain a solution

$$u = \phi(x,y) = g(x,y,\psi(x,y))$$

in some neighborhood of (x_0, y_0).

The functions $u = \phi(x,y)$ and $v = \psi(x,y)$ are solutions to the system of equations

$$\begin{aligned} F(x,y,u,v) &= 0 \\ G(x,y,u,v) &= 0 \end{aligned}$$

for (x,y) in some neighborhood I_0 of (x_0,y_0) and (u,v) in some neighborhood J_0 of (u_0,v_0). From Theorem 12.44 we are assured that the functions ϕ and ψ have continuous first partials. It remains to verify the formulas stated for these partials.

To check the formula for ϕ_1, recall that $u = \phi(x,y)$ is a solution of the equation $H(x,y,u) = 0$. Using (92), we obtain

$$\begin{aligned} \phi_1 &= -\frac{H_1}{H_3} = -\frac{G_1 + G_4 g_1}{-\frac{|J|}{F_4}} \\ &= -\frac{G_1 - G_4\left(-\frac{F_1}{F_4}\right)}{-\frac{|J|}{F_4}} \\ &= -\frac{F_4 G_1 - G_4 F_1}{|J|} = -\frac{1}{|J|}\frac{\partial(F,G)}{\partial(x,v)}. \end{aligned}$$

Formulas for the other partials are obtained similarly. ■

Note. We stated Theorem 12.48 for two equations in four variables. Theorem 12.44 of the previous section deals with the case of one equation in several variables. A general theorem involving m equations in $n+m$ variables follows similar lines. A typical one of the m equations is of the form

$$F^i(x_1, \ldots, x_n, u_1, \ldots, u_m) = 0.$$

(Here, the superscript identifies the function so as not to confuse the equation number with a partial derivative, which might occur if subscripts were used.) We assume that F^i has continuous partial derivatives in some region in $\mathbb{R}^{m+n}$ and that the appropriate Jacobian determinant,

$$\frac{\partial(F^1, \ldots, F^m)}{\partial(u_1, \ldots, u_m)},$$

is not zero. The conclusions are then the obvious analogues of the conclusion to Theorem 12.48.

Exercises

12.6.7 Verify those formulas not verified in the text for the partial derivatives in Theorem 12.48.

12.6.4 Inverse Function Theorem

We now apply Theorem 12.48 to obtain a theorem concerning inverses of mappings. Suppose we have equations of the form

$$\begin{aligned} x &= f(u,v) \\ y &= g(u,v), \end{aligned}$$

where f and g are defined on an open set $D \subset \mathbb{R}^2$. These equations determine a function $T : D \to \mathbb{R}^2$, $T(u,v) = (x,y)$. We often call such functions *mappings* or *transformations*. If we can solve these equations for u and v for values of (x,y) in some set $D' \subset \mathbb{R}^2$,

$$\begin{aligned} u &= \phi(x,y) \\ v &= \psi(x,y), \end{aligned} \tag{93}$$

then this determines a mapping $S : D' \to \mathbb{R}^2$ such that $T \circ S$ is the identity on D'. Thus S and T are, in some sense, inverses of each other. Some care must be taken in making this inverse relationship between S and T precise. The problem has to do with the domains of S and T on which $S \circ T$ and $T \circ S$ are the identity.

Example 12.49: Consider the function of one variable

$$u = f(x) = x^2$$

defined on $\mathbb{R}$. If we let $g(u) = \sqrt{u} \quad (u \geq 0)$, then

$$(f \circ g)(u) = (\sqrt{u})^2 = u,$$

so $f \circ g$ is the identity on the set $\{u : u \geq 0\}$. But

$$(g \circ f)(x) = \sqrt{x^2} = |x|,$$

so $g \circ f$ is the identity on the set $\{x : x \geq 0\}$ but not on all of $\mathbb{R}$. Since f is not one-to-one on $\mathbb{R}$, its domain must be reduced to a smaller one on which f is one-to-one. When this is done properly, f does have an inverse. If we had chosen a specific value of $x_0 \neq 0$, we would have been able to find a neighborhood I of x_0 and a neighborhood J of $u_0 = f(x_0)$ such that $f(I) = J$ and f is one-to-one on I. ◀

Theorem 12.50 provides conditions under which such a local inverse exists for mappings from $\mathbb{R}^2 \to \mathbb{R}^2$.

Theorem 12.50 (Inverse Function Theorem) *Let f and g have continuous first partial derivatives on an open set $D \subset \mathbb{R}^2$. Let $(u_0, v_0) \in D$ and suppose*

$$x_0 = f(u_0, v_0)\ ,\ y_0 = g(u_0, v_0).$$

Suppose further that

$$|J| = \frac{\partial(f,g)}{\partial(u,v)} \neq 0 \text{ at } (u_0, v_0).$$

Then there are neighborhoods I_0 of (x_0, y_0) and J_0 of (u_0, v_0) such that for each $(x,y) \in I_0$ there corresponds a unique $(u,v) \in J_0$ with

$$x = f(u,v)\ ,\ y = g(u,v).$$

This correspondence determines functions from I_0 into J_0

$$u = \phi(x,y)\,,\ v = \psi(x,y).$$

The functions ϕ and ψ have continuous partial derivatives given by

$$\begin{aligned}\frac{\partial\phi}{\partial x} &= -\frac{1}{|J|}\frac{\partial g}{\partial v}, \quad \frac{\partial\phi}{\partial y} = -\frac{1}{|J|}\frac{\partial f}{\partial v},\\ \frac{\partial\psi}{\partial x} &= -\frac{1}{|J|}\frac{\partial g}{\partial u}, \quad \frac{\partial\psi}{\partial y} = -\frac{1}{|J|}\frac{\partial f}{\partial u}.\end{aligned}$$

Proof. Theorem 12.50 is just a special case of Theorem 12.48. Let

$$\begin{aligned}F(x,y,u,v) &= f(u,v)-x\\ G(x,y,u,v) &= g(u,v)-y.\end{aligned}$$

Applying Theorem 12.48, we obtain the functions ϕ and ψ and the formulas for the partial derivatives. For example, to obtain $\frac{\partial\psi}{\partial y}$ we need only calculate, using Theorem 12.48,

$$\begin{aligned}\frac{\partial\psi}{\partial y} &= -\frac{1}{|J|}\frac{\partial(F,G)}{\partial(u,x)} = -\frac{1}{|J|}\begin{vmatrix}\frac{\partial F}{\partial u} & \frac{\partial F}{\partial x}\\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial x}\end{vmatrix}\\ &= -\frac{1}{|J|}\begin{vmatrix}\frac{\partial f}{\partial u} & -1\\ \frac{\partial f}{\partial u} & 0\end{vmatrix} = -\frac{1}{|J|}\frac{\partial f}{\partial u}.\end{aligned}$$

The other calculations are similar. ■

Example 12.51: Consider the equations in Example 12.46.

$$\begin{aligned}x &= r\cos\theta\\ y &= r\sin\theta\end{aligned}$$

The Jacobian determinant is

$$\begin{vmatrix}\cos\theta & -r\sin\theta\\ \sin\theta & r\cos\theta\end{vmatrix} = r(\cos^2\theta+\sin^2\theta) = r.$$

For $(r_0,\theta_0)\in\mathbb{R}^2$, $(r_0\neq 0)$ there will be a neighborhood of $(r_0,\theta_0)\in\mathbb{R}^2$ on which an inverse to the transformation exists. There is no inverse on all of $\mathbb{R}^2$ or even on any D containing both of the points $(4,0)$ and $(4,2\pi)$, because

$$(4\cos 0, 4\sin 0) = (4\cos 2\pi, 4\sin 2\pi) = (4,0).$$

At these points the Jacobian determinant does not equal zero, but the transformation is not one-to-one on D. ◀

Exercises

12.6.8 The standard formulas relating spherical to rectangular coordinates are

$$z=\rho\sin\phi\cos\theta,\ y=\rho\sin\theta,\ z=\rho\cos\phi.$$

Can these equations be solved for ρ, ϕ, and θ in terms of x, y, and z?

12.6.9 State in full detail the analogue of Theorem 12.48 arising from four equations in the six variables w,x,y,z,s,t that involve solving for s and t in terms of the other variables.

12.6.10 Verify those formulas not verified in the text for the partial derivatives in Theorem 12.50.

12.6.11 Does the pair of equations $x = u^2 - v^2$, $y = 2uv$, have an inverse on a neighborhood of $(0,0)$?

12.6.12 Does the pair of equations $x = r\cos\theta$, $y = r\sin\theta$, have an inverse on a neighborhood of $(0,0)$?

12.6.13 Show that the Jacobian determinant of the mapping

$$x = e^u \cos v, \; y = e^u \sin v$$

is never zero, yet the mapping is not one-to-one on all of $\mathbb{R}^2$.

12.6.14 Under the hypotheses of Theorem 12.50 show

$$\frac{\partial(u,v)}{\partial(x,y)} = \left[\frac{\partial(x,y)}{\partial(u,v)}\right]^{-1}.$$

12.6.15 Let f and g have continuous first partial derivatives on an open set $D \subset \mathbb{R}^2$ and let $T : D \to \mathbb{R}^2$ be defined by

$$T(u,v) = (f(u,v), g(u,v)).$$

Prove that if T is one-to-one on D, then the set $T(D)$ is open.

12.7 Functions From $\mathbb{R} \to \mathbb{R}^m$

We shall be concerned with differentiability of a function f defined on a subset of $\mathbb{R}$ with values in $\mathbb{R}^m$, $m > 1$. Suppose we are given two equations

$$\begin{aligned} x &= x(t) \\ y &= y(t) \end{aligned} \qquad (a \le t \le b). \tag{94}$$

These equations determine a function $f : [a,b] \to \mathbb{R}^2$ defined by

$$f(t) = (x(t), y(t)).$$

In elementary calculus we often view the equations (94) as a parametric representation of a curve. As t moves from a to b, $f(t)$ traces out a curve. Students learn that, under appropriate hypotheses, we can determine the slope of the tangent line at a point $(x(t), y(t))$ by calculating

$$\frac{dy}{dx} = \frac{y'(t)}{x'(t)}.$$

In this section we view the function f simply as a function from $[a,b]$ to $\mathbb{R}^2$. We shall define differentiability of f in terms of approximations of f by a linear function. Our requirement for differentiability at t is *not* the same as the requirement that the curve given parametrically by (94) has a tangent at $(x(t), y(t))$. (Indeed, Example 12.52 will show that a curve need not have such a tangent even when f is differentiable at t.)

How then should we define differentiability of f? We take our cue from material in preceding sections. We wish to approximate a function f near a point t by a linear function T. The approximation should be good in the sense that $\|f(t) - T(t)\|$ is small in comparison with the distance between t and t_0 when t is near t_0. Since

$$T(t_0 + h) - T(t_0) = T(h)$$

for linear functions, we can write this in the form

$$\lim_{h\to 0} \frac{\|f(t_0+h)-f(t_0)-T(h)\|}{h} = 0. \tag{95}$$

Let's look at the familiar setting, $m = 2$, that we used as an introduction to this section:

$$\begin{aligned} x &= x(t) \\ y &= y(t) \end{aligned} \quad (a \le t \le b),$$

where x and y are continuous functions on $[a,b]$. As we mentioned, these equations define a function $f : [a,b] \to \mathbb{R}^2$ by

$$f(t) = (x(t), y(t)).$$

We would like to define differentiability of f at a point $t \in [a,b]$ according to a criterion similar to (95).

Suppose for the moment that x and y are differentiable at t. Let

$$T(h) = (x'(t), y'(t))h.$$

Then $f(t+h) - f(t) - T(h)$ becomes

$$\begin{gathered}(x(t+h)-x(t), y(t+h)-y(t)) - (x'(t)h, y'(t)h) = \\ (x(t+h)-x(t)-x'(t)h, y(t+h)-y(t)-y'(t)h).\end{gathered}$$

In looking at these expressions, several things are immediately apparent:

1. $f(t+h) - f(t) - T(h)$ is a vector in $\mathbb{R}^2$, not a real number, so the numerator in (95) required a norm in place of the absolute value.

2. $T(h) = (x'(t), y'(t))h$ is linear in h, as required. This means that, for all α, h_1, $h_2 \in \mathbb{R}$,

   $$T(\alpha h) = \alpha T(h)$$

   and

   $$T(h_1 + h_2) = T(h_1) + T(h_2).$$

   (The point t is fixed in this discussion.)

3. Since x and y are differentiable (by assumption), then, as $h \to 0$,

   $$\begin{aligned} \frac{x(t+h)-x(t)-x'(t)h}{h} &\to 0, \\ \frac{y(t+h)-y(t)-y'(t)h}{h} &\to 0, \end{aligned}$$

   and hence

   $$\frac{\|f(t+h)-f(t)-T(h)\|}{|h|} \to 0.$$

More generally, we obtain the following. Let f be defined on a neighborhood of a point $t \in \mathbb{R}$ with range in $\mathbb{R}^m$. We can write

$$f(t) = (x_1(t), \ldots, x_m(t)),$$

where $x_1,\dots,x_m$ are all defined on a neighborhood of t. If each of the functions $x_1,\dots,x_m$ is differentiable at t, then

$$\lim_{h\to 0}\frac{\|f(t+h)-f(t)-T(h)\|}{h}=0, \tag{96}$$

where

$$T(h)=(x_1'(t),\dots,x_m'(t))h. \tag{97}$$

The function T is linear in h. (See Exercise 12.7.1.) We say f is *differentiable* at t and that the *differential* of f at t is the function T whose value at h is $T(h)$. Note that T depends on the point t.

Note. Our definition of differentiability at t assumes the differentiability of each of the functions $x_1,\dots,x_m$ at t. This assumption is similar to our definition for real-valued functions of several variables in which we assumed existence of the appropriate partial derivatives. This assumption may seem natural, but technically we are seeking *any* linear function T that satisfies (96). A priori, it need not satisfy (97).

It is a fact, however, that if f is differentiable at t, then the functions $x_1,\dots,x_m$ are differentiable at t. Furthermore, if a linear function T satisfies (96), it must be given by (97).

To see this, suppose $f:\mathbb{R}\to\mathbb{R}^m$ is differentiable at t and $T=T_t$ is a linear function of h. Then there exists $(a_1,\dots,a_m)\in\mathbb{R}^m$ such that $T(h)=(a_1,\dots,a_m)h$ for all $h\in\mathbb{R}$. The ith component of the vector

$$\frac{f(t+h)-f(t)-T(h)}{h}$$

is

$$\frac{x_i(t+h)-x_i(t)}{h}-a_i.$$

In order for (96) to hold, it is necessary and sufficient that

$$\lim_{h\to 0}\frac{x_i(t+h)-x_i(t)}{h}=a_i$$

for $i=1,\dots,m$, that is, that $a_i=x_i'(t)$, as was to be shown.

At this point we should contrast two statements that students sometimes confuse. Both statements are true.

> *A function $f:\mathbb{R}\to\mathbb{R}^m$ is differentiable at t if and only if all the coordinate functions $x_1,\dots,x_m$ are differentiable at t.*
>
> *If $f:\mathbb{R}^n\to\mathbb{R}$ is differentiable at $x\in\mathbb{R}^n$, then all the partials $\dfrac{\partial f}{\partial x_1},\dots,\dfrac{\partial f}{\partial x_n}$ exist at x. The converse is false: All the partials can exist without f being differentiable.*

The situation is similar to a corresponding one for continuity. $f:\mathbb{R}\to\mathbb{R}^m$ is continuous at a point t if and only if all the coordinate functions are continuous at the point. But a functions $f:\mathbb{R}^n\to\mathbb{R}$ can be continuous in each variable separately at a point without being continuous there.

We end this section by noting that the reason for taking our definition of differentiability was to obtain the kind of linear approximation to f that we wanted—approximations consistent with the spirit of those in previous sections. It does *not* imply that the curve given parametrically by

$$\begin{array}{rcl} x_1 &=& x_1(t) \\ x_2 &=& x_2(t) \\ \vdots & & \vdots \\ x_m &=& x_m(t) \end{array} \qquad a \le t \le b$$

has a tangent line at a point $(x_1(t),\ldots,x_m(t))$ just because all the $x_1,\ldots,x_m$ are differentiable at t.

Example 12.52: Define the curve

$$\begin{aligned} x(t) &= \begin{cases} t^2, & \text{if } 0 \le t \le 1 \\ -t^2, & \text{if } -1 \le t < 0 \end{cases} \\ y(t) &= t^2, \quad \text{if } -1 \le t \le 1. \end{aligned} \tag{98}$$

Then $x'(0) = y'(0) = 0$ so the function $f(t) = (x(t), y(t))$ is differentiable at $t = 0$. But the curve given by (98) is just the curve $y = |x|$ which does not have a tangent at $(0,0) = f(0)$. ◀

Exercises

12.7.1 Show that T, as given in (97) is linear in h, that is, $T(\alpha h) = \alpha T(h)$ and $T(h_1 + h_2) = T(h_1) + T(h_2)$ whenever α, h_1, $h_2 \in \mathbb{R}$.

12.7.2 Let

$$\begin{array}{rcl} x(t) &=& t \\ y(t) &=& |t| \end{array} \qquad (-1 \le t \le 1).$$

Show that these equations define the same curve as the curve given by (98), yet the function $g(t) = (x(t), y(t))$ given here is not differentiable at $t = 0$. Show that there is *no* linear function T that satisfies equation (96).

12.8 Functions From $\mathbb{R}^n \to \mathbb{R}^m$

✂ *Advanced section. May be omitted.*

✂ This section requires some familiarity with linear algebra and may be omitted if necessary.

We have discussed thus far differentiability of functions from $\mathbb{R}^n$ to $\mathbb{R}$ and from $\mathbb{R}$ to $\mathbb{R}^m$. These are, of course, just special cases of the more general setting, functions from $\mathbb{R}^n$ to $\mathbb{R}^m$. We now turn to this general setting. Indeed much of what we covered so far was in an attempt to get to this level of abstraction in a gentle manner.

We first recall that our objective in Sections 12.4 and 12.7 was to approximate a function by a linear function in a certain sense. We have a similar objective here. The natural exposition of the material involves a linear algebraic viewpoint.

12.8.1 Review of Differentials and Derivatives

✂ *Enrichment section. May be omitted.*

Let's review our notion of the differential from the earlier settings but now using some language from linear algebra. For a differentiable function

$$f : \mathbb{R}^n \to \mathbb{R}$$

the differential at a point $x = (x_1, \ldots, x_n)$ took the form

$$df(x,h) = \frac{\partial f}{\partial x_1} h_1 + \cdots + \frac{\partial f}{\partial x_n} h_n,$$

where $h = (h_1, \ldots, h_n)$ and the partials are evaluated at x. We can view this as a product of the $1 \times n$ matrix

$$\left(\frac{\partial f}{\partial x_1}, \ldots . \frac{\partial f}{\partial x_n} \right)$$

with the $n \times 1$ matrix

$$\begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix},$$

so that the differential is the product of a row vector with a column vector. If we replace h_i by dx_i, $i = 1, \ldots, n$, in our notation and we define f' by

$$f'(x) = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right),$$

then we can write

$$d(f, dx) = f'(x)dx,$$

where

$$dx = (dx_1, \ldots, dx_n).$$

(This notation is reminiscent of the notation in elementary calculus for the differential of a function from $\mathbb{R}$ to $\mathbb{R}$.)

Let us now look at the situation we discussed in Section 12.7, where we dealt with functions $f : \mathbb{R} \to \mathbb{R}^m$ so that, in our usual notation,

$$f = (f^1, f^2, f^3, \ldots, f^m).$$

There the differential took the form

$$df(x,h) = \left(\frac{df^1}{dx}, \frac{df^2}{dx}, \ldots, \frac{df^m}{dx} \right) h,$$

with the derivatives evaluated at x. We can write this in matrix notation as

$$df(x,h) = \begin{pmatrix} \frac{df^1}{dx} \\ \frac{df^2}{dx} \\ \vdots \\ \frac{df^m}{dx} \end{pmatrix} (h) \quad .$$

Domain	Range	$A = f'(x) =$	$df(x,h) = f'(x)h$
$\mathbb{R}$	$\mathbb{R}$	$f'(x)$	$f'(x)h$
$\mathbb{R}^n$	$\mathbb{R}$	$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_1}\right)$	$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}$
$\mathbb{R}$	$\mathbb{R}^m$	$\begin{pmatrix} \frac{df^1}{dx} \\ \frac{df^2}{dx} \\ \vdots \\ \frac{df^m}{dx} \end{pmatrix}$	$\begin{pmatrix} \frac{df^1}{dx} \\ \frac{df^2}{dx} \\ \vdots \\ \frac{df^m}{dx} \end{pmatrix} (h)$

Figure 12.8. The form of the derivative and differential.

If we write

$$f'(x) = \begin{pmatrix} \frac{df^1}{dx} \\ \frac{df^2}{dx} \\ \vdots \\ \frac{df^m}{dx} \end{pmatrix},$$

then the differential takes the form

$$df(x,h) = f'(x)h \text{ or } df(x,dx) = f'(x)\,dx.$$

In both of these cases, we have a notion of a derivative $f'(x)$ and of a differential

$$df(x,h) = f'(x)h$$

or

$$df(x,dx) = f'(x)dx.$$

In each case we can view the derivative $f'(x)$ as a matrix representing a certain linear transformation, A_x, and the differential as the result of applying this transformation to the vector h:

$$df(x,h) = A_x(h).$$

Summary Let us summarize the preceding with a chart. We have a function defined on an open set D in $\mathbb{R}$ or in $\mathbb{R}^n$, with range in $\mathbb{R}$ or $\mathbb{R}^m$. The table in Figure 12.8.1 indicates the form of the derivative and of the differential at point x.

Note. We view $f'(x)$ and h as matrices representing linear transformations, and $df(x,h)$ as a product of these matrices. (When we are dealing with the case of a real function on $\mathbb{R}$, the product of two numbers can be viewed as the product of 1×1 matrices.)

12.8.2 Definition of the Derivative

✂ *Enrichment section. May be omitted.*

Let us now turn to the general case. Suppose we have m linear equations in n variables

$$\begin{array}{rcl} y_1 &=& a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ y_2 &=& a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots & & \\ y_m &=& a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{array} \tag{99}$$

These equations define a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. We can write these equations as a matrix equation

$$y = Ax,$$

where $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_m)$, and $A = (a_{ij})$ is the $m \times n$ matrix of coefficients.

A general function from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be expressed in a form similar to (99). For $f = (f^1, \ldots, f^m)$ write

$$\begin{array}{rcl} y_1 &=& f^1(x_1, \ldots, x_n) \\ y_2 &=& f^2(x_1, \ldots, x_n) \\ \vdots & & \\ y_m &=& f^m(x_1, \ldots, x_n) \end{array} . \tag{100}$$

(Here, as before, superscripts are used to identify the functions so as not to confuse them with partial derivatives.)

Our task is to approximate a function given by (100) using a linear transformation given by (99). Specifically, we seek a linear transformation T from $\mathbb{R}^n$ to $\mathbb{R}^m$, with $T = T_x$ depending on x, such that

$$\lim_{\|h\| \to 0} \frac{\|f(x+h) - f(x) - T(h)\|}{\|h\|} = 0.$$

The norms involved can be any of the norms for $\mathbb{R}^m$ for the numerator and any of the norms for $\mathbb{R}^n$ for the denominator $\|h\|$. This is so because, as we saw in Section 11.6, convergence with respect to one norm is equivalent to convergence with respect to another norm. To be specific, however, we shall use $\|t\| = \|t\|_1 =$ the sum of the absolute values of the components of t.

Definition 12.53: Let D be an open set in $\mathbb{R}^n$, x a point in D and suppose

$$f : D \to \mathbb{R}^m.$$

We say f is *differentiable* at x if there exists a linear transformation T_x from $\mathbb{R}^n$ to $\mathbb{R}^m$ such that

$$\lim_{\|h\| \to 0} \frac{\|f(x+h) - f(x) - T_x(h)\|}{\|h\|} = 0. \tag{101}$$

Definition 12.54: The *differential* of f, denoted by df is defined at x by

$$df(x,h) = T_x(h)$$

for all $h \in \mathbb{R}^n$.

Definition 12.55: The *derivative* of f at x is the function

$$f'(x) = T_x$$

defined on the set of points of differentiability of f.

Observe that $f'(x)$ is the linear transformation T_x. Its domain is $\mathbb{R}^n$ and its range is $\mathbb{R}^m$. When f is differentiable at x we can write (101) in the form

$$\lim_{\|h\|\to 0} \frac{\|f(x+h) - f(x) - f'(x)h\|}{\|h\|} = 0. \tag{102}$$

This form resembles the familiar form of differentiability for functions from $\mathbb{R}$ to $\mathbb{R}$, when absolute values are replaced by norms.

Exercises

12.8.1 Verify that the validity of (101) does not depend on the norms for $\mathbb{R}^n$ and $\mathbb{R}^m$ that are used.

12.8.2 Let A be a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. Show that for each $x \in \mathbb{R}^n$, $A'(x) = A$.
SEE NOTE 285

12.8.3 Suppose $f, g : \mathbb{R}^n \to \mathbb{R}^m$ and $\alpha \in \mathbb{R}$. Prove that if f and g are differentiable at x, then so too are $f+g$ and αf. In addition,

$$(f+g)'(x) = f'(x) + g'(x)$$

and

$$(\alpha f)'(x) = \alpha f'(x).$$

12.8.4 Suppose $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^m$. Prove that if f and g are differentiable at x, then so too is the product function fg and that

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

12.8.3 Jacobians

✂ *Enrichment section. May be omitted.*

With these preliminaries completed, we ask our first question. If f is differentiable at x, can we represent the linear transformation $f'(x)$ by a matrix involving partial derivatives? For the cases of mappings from $\mathbb{R}^n$ to $\mathbb{R}$ or from $\mathbb{R}$ to $\mathbb{R}^m$, we saw how to do this (Fig. 12.8.1). These results suggest that a matrix representation of $f'(x)$ would be an $m \times n$ Jacobian matrix. This is indeed the case.

Theorem 12.56: *Let D be an open subset of $\mathbb{R}^n$ and let $f : D \to \mathbb{R}^m$. Let $(f^1, \ldots, f^m)$ be the component functions of f as in (100). Thus*

$$f = (f^1, \ldots, f^m)$$

with $f^i : D \to \mathbb{R}$ for each $i = 1, \ldots, m$. Then f is differentiable at $x \in D$ if and only if each of the functions f^i is differentiable at x. When f is differentiable at x, the derivative

$f'(x)$ can be represented by the Jacobian matrix

$$T_x = f'(x) = \begin{pmatrix} \frac{\partial f^1}{\partial x_1} & \frac{\partial f^1}{\partial x_2} & \cdots & \frac{\partial f^1}{\partial x_n} \\ \frac{\partial f^2}{\partial x_1} & \frac{\partial f^2}{\partial x_2} & \cdots & \frac{\partial f^2}{\partial x_n} \\ \vdots & & & \\ \frac{\partial f^m}{\partial x_1} & \frac{\partial f^m}{\partial x_2} & \cdots & \frac{\partial f^m}{\partial x_n} \end{pmatrix}, \tag{103}$$

where the partial derivatives are evaluated at x.

Proof. The proof of this theorem involves no more than looking at the component functions f^i. Suppose first that f is differentiable at x. This means that

$$\lim_{\|h\|\to 0} \frac{\|f(x+h) - f(x) - f'(x)h\|}{\|h\|} = 0. \tag{104}$$

Now $f'(x)$, being a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$, has a matrix representation $A = (a_{ij})$. Thus if $h = (h_1, \ldots, h_n)$, then the ith component of $f'(x)h$ is $\sum_{j=1}^{n} a_{ij}h_j$, and the ith component of

$$f(x+h) - f(x)$$

is $f^i(x+h) - f^i(x)$. As a result, the ith component of

$$f(x+h) - f(x) - f'(x)h$$

is

$$f^i(x+h) - f^i(x) - \sum_{j=1}^{n} a_{ij}h_j.$$

Now

$$|f^i(x+h) - f^i(x) - \sum_{j=1}^{n} a_{ij}h_j| \le \|f(x+h) - f(x) - f'(x)h\|$$

so, because of (104),

$$\lim_{\|h\to 0\|} \frac{|f^i(x+h) - f^i(x) - \sum_{j=1}^{n} a_{ij}h_j|}{\|h\|} = 0.$$

Thus f^i is differentiable at x. Furthermore,

$$a_{ij} = \frac{\partial f^i}{\partial x_j}. \tag{105}$$

The verification of (105) is similar to the proof of the corresponding fact in the previous section. (We leave it as Exercise 12.8.5.)

To prove the converse, assume each of the functions f^i is differentiable at x. This means

$$\lim_{\|h\|\to 0} \frac{|f^i(x+h) - f^i(x) - \sum_{j=1}^{n} \frac{\partial f^i}{\partial x_j}h_j|}{\|h\|} = 0. \tag{106}$$

Now (106) is valid for each $i = 1, \ldots, m$. But the norm in the numerator of (104) is just the sum

$$\sum_{i=1}^{m} \left| f^i(x+h) - f^i(x) - \sum_{j=1}^{n} \frac{\partial f^i}{\partial x_j} h_j \right|,$$

each term of which approaches zero when divided by $\|h\|$. Hence so does the sum, and (104) is satisfied. Thus f is differentiable at x. ■

Observe that the requirement for differentiability of f at x can be written in the form

$$f(x+h) - f(x) = f'(x)h + E(h)\|h\|, \tag{107}$$

where the error term E satisfies

$$\lim_{\|h\| \to 0} \|E(h)\| = 0 \text{ and } E(0) = 0.$$

This observation shows that when f is differentiable at x, f is also continuous at x. It also highlights the sense in which the linear transformation $f'(x)$ approximates f near x. We see that (107) is just a compact way of expressing condition 2 of Definition 12.16 when we are dealing with functions from $\mathbb{R}^n$ to $\mathbb{R}$.

Example 12.57: An example will clarify (107). Let

$$\begin{aligned} u &= xy \\ v &= x+y \\ f(x,y) &= (u,v). \end{aligned}$$

By (103)

$$f'(x) = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} y & x \\ 1 & 1 \end{pmatrix}.$$

Then (107) becomes

$$\begin{aligned} &((x+h)(y+k) - xy, (x+h) + (y+k) - (x+y)) = \\ &\begin{pmatrix} y & x \\ 1 & 1 \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix} + E(h,k)(|h| + |k|). \end{aligned}$$

Simplifying and solving for $E(h,k)$ gives

$$E(h,k) = \frac{(hk, 0)}{|h| + |k|}.$$

Thus

$$\|E(h,k)\| = \frac{|hk|}{|h| + |k|}.$$

If $h = 0$ or $k = 0$, $\|E(h,k)\| = 0$. Otherwise,

$$\|E(h,k)\| = \frac{1}{|1/k| + |1/h|}.$$

Thus $\lim_{(h,k) \to (0,0)} \|E(h,k)\| = 0$. ◄

Exercises

12.8.5 Verify equation (105) in the proof of Theorem 12.56.

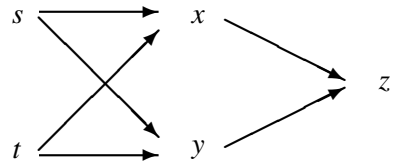
12.8.4 Chain Rules

✂ *Enrichment section. May be omitted.*

In Section 12.5 we discussed chain rules and we indicated how employing a "schema" might be useful in writing down relevant chains. All such chain rules, as well as the familiar calculus chain rule (Theorem 7.11) involving functions from $\mathbb{R}$ to $\mathbb{R}$, are special cases of a general chain rule. We consider this chain rule next. Our framework of linear algebra together with the convenient notation we adopted in this section make this chain rule easy to prove and easy to remember.

Before we state and prove this chain rule, let us recall one of the chain rules (Theorem 12.36) from Section 12.5.6. This is the rule associated with the schema we have already seen in Example 12.30:

(108)



In the terminology of this section, we have here two functions, $f : \mathbb{R}^2 \to \mathbb{R}^2$, with coordinate functions x and y; and $g : \mathbb{R}^2 \to \mathbb{R}$ indicated in (108) as z, $z = g(x,y)$. Assuming differentiability as needed, ignoring for the moment the points at which partials are evaluated, and using the notation suggested by (108), we find g' represented by the matrix $B = \left(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right)$ and f' by the matrix

$$A = \begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{pmatrix}.$$

Thus

$$\begin{aligned} BA &= \left(\tfrac{\partial z}{\partial x}, \tfrac{\partial z}{\partial y}\right) \begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{pmatrix} \\ &= \left(\tfrac{\partial z}{\partial x}\tfrac{\partial x}{\partial s} + \tfrac{\partial z}{\partial y}\tfrac{\partial y}{\partial s}, \tfrac{\partial z}{\partial x}\tfrac{\partial x}{\partial t} + \tfrac{\partial z}{\partial y}\tfrac{\partial y}{\partial t}\right). \end{aligned}$$

This last expression is just the derivative of the composite function $F = g \circ f$, since $F' = \left(\frac{\partial z}{\partial s}, \frac{\partial z}{\partial t}\right)$ and the components of F' are by Example 12.30 (or Theorem 12.36) the components of the vector BA.

The preceding discussion is not precise, but it does suggest a chain rule that concludes with the appealing formula

$$(g \circ f)'(x) = g'(f(x))f'(x).$$

Before we attempt a proof of this chain rule formula (which we do in Theorem 12.58), let us try to get a sense of why it works. In Section 7.3.2 we motivated the chain rule for functions from $\mathbb{R}$ to $\mathbb{R}$ by viewing the absolute value of the derivative at a point

as a local magnification factor. When dealing with $f : \mathbb{R}^n \to \mathbb{R}^n$ we can do something similar by replacing the absolute value of f' by the determinant of the Jacobian.

Suppose first that A is a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$, say

$$A(u,v) = (x,y) \quad (x = au,\ y = bv).$$

Then

$$A'(u,v) = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

If we view A as a mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$, we see it stretches a set E horizontally by a factor of a and vertically by a factor of b. (For example, if $a = 2$ and $b = 3$, A maps the unit square $[0,1] \times [0,1]$ onto the rectangle $[0,2] \times [0,3]$.) Thus if E and $A(E)$ have area,

$$\frac{\text{Area}\, A(E)}{\text{Area}\, E} = ab.$$

The mapping A magnifies the area of a set by a factor of ab. This is reflected by calculating the determinant of A',

$$\begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab.$$

Now take any $f : \mathbb{R}^2 \to \mathbb{R}^2$ that is differentiable at a point $(u,v) \in \mathbb{R}^2$. If

$$f'(u,v) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix},$$

then the mapping defined by f has a magnification factor ab in a limit sense near (u,v). In suggestive notation similar to that in Section 7.3.2 we could write

$$\lim_{J \to (u,v)} \frac{\text{Area}\, f(J)}{\text{Area}\, J} = ab.$$

This is so because of the sense in which the linear transformation approximates f near a point of differentiability of f.

Now consider another mapping $g : \mathbb{R}^2 \to \mathbb{R}^2$ with derivative g' at $f(x)$. Its derivative at $f(x)$ is a linear transformation $B = g'(f(x))$. The magnification factor of g at $f(x)$ is the same as the magnification factor of B, namely the determinant of B.

Thus if α is the magnification factor of A (and of f at x) and β is the magnification factor of B (and of g at $f(x)$), then $\beta\alpha$ is the magnification factor of BA (and of $g \circ f$ at x). This suggests that the linear transformation BA is the derivative of $g \circ f$ at x, at least for the case $n = m = p$. Theorem 12.58 says this is the case for all m, n, p.

We need to recall a bit of linear algebra that will appear in the proof of Theorem 12.58. Let A be a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. Since A and the norm are continuous (see Exercise 11.7.10), $\|Ax\|$ achieves a maximum on the compact set $\{x : \|x\| = 1\}$. We denote this maximum by $\|A\|$. Thus

$$\|A\| = \max_{\|x\|=1} \|Ax\|.$$

We use this "norm" notation because $\|\cdot\|$ is actually a norm on the set of linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$, but we will not need this fact. Observe that for all $x \in \mathbb{R}^n$,

($x \neq 0$),

$$\|Ax\| = \left\| A\left(\frac{x}{\|x\|}\right)\right\| \|x\| \leq \|A\|\,\|x\|.$$

In the proof that follows there will be several different norms, all denoted by $\|\cdot\|$. Each of the spaces $\mathbb{R}^n$, $\mathbb{R}^m$,‘ and $\mathbb{R}^p$ have norms, as do the sets of linear transformations from one of the spaces to another. We could have used different notations for different norms, (e.g., $\|\|\cdot\|\|$) but this might cause eye strain. Instead, we follow the common practice of using only one symbol for all the norms, trusting that the context will make it clear which norm is meant.

Exercises

12.8.6 Show that the set of all linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a vector space (under appropriate interpretations of sum and scalar product) and show that

$$\|A\| = \max_{\|x\|=1} \|Ax\|$$

defines a norm for elements of this space.

12.8.5 Proof of Chain Rule

✂ *Enrichment section. May be omitted.*

We are now in a position to give a formal statement and proof of the general chain rule formula.

Theorem 12.58: *Let D be an open set in $\mathbb{R}^n$. Let $f : D \to \mathbb{R}^m$ be differentiable at $x_0 \in D$. Let H be an open set containing $f(D)$ and let $g : H \to \mathbb{R}^p$ be differentiable at $f(x_0)$. Then the function $F : D \to \mathbb{R}^p$ defined by*

$$F(x) = g(f(x))$$

is differentiable at x_0 and

$$F'(x_0) = g'(f(x_0))f'(x_0). \tag{109}$$

The product in (109) is a product of two linear transformations.

Proof. Let

$$y_0 = f(x_0),\ \ A = f'(x_0),\ \text{ and }\ B = g'(y_0).$$

We wish to show that

$$\frac{\|F(x_0+h) - F(x_0) - BAh\|}{\|h\|} \to 0 \text{ as } h \to 0. \tag{110}$$

Since f is differentiable at x_0 and g is differentiable at y_0 we can, using (107), write for $x_0 + h \in D$ and $y_0 + k \in H$,

$$\begin{array}{rcl} f(x_0+h) - f(x_0) - Ah &=& E_1(h)\|h\|, \\ g(y_0+k) - g(y_0) - Bk &=& E_2(k)\|k\| \end{array}, \tag{111}$$

where

$$\|E_1(h)\| \to 0 \text{ as } h \to 0 \text{ and } \|E_2(k)\| \to 0 \text{ as } k \to 0. \tag{112}$$

We now estimate the numerator in (110). For a given h, let

$$k = f(x_0 + h) - f(x_0). \tag{113}$$

From (111) we obtain

$$\begin{aligned} \|k\| &= \|A\,h + E_1(h)\,\|h\|\,\| \\ &\leq \|A\|\,\|h\| + \|E_1(h)\|\,\|h\| \\ &= (\|A\| + \|E_1(h)\|)\,\|h\|. \end{aligned} \tag{114}$$

From (113) we see that

$$f(x_0 + h) = f(x_0) + k = y_0 + k,$$

so

$$\begin{aligned} &F(x_0 + h) - F(x_0) = \\ &\quad g(f(x_0 + h)) - g(y_0) = g(y_0 + k) - g(y_0). \end{aligned}$$

Thus

$$F(x_0 + h) - F(x_0) - BAh = g(y_0 + k) - g(y_0) - BAh. \tag{115}$$

Using (111), we express the right side of (115) as

$$\begin{aligned} &Bk + E_2(k)\,\|k\| - BAh = \\ &\quad B(k - Ah) + E_2(k)\,\|k\| = BE_1(h)\,\|h\| + E_2(k)\,\|k\|. \end{aligned}$$

This estimate for

$$F(x_0 + h) - F(x_0) - BAh$$

together with (112) and (114), implies that for $h \neq 0$,

$$\begin{aligned} &\frac{\|F(x_0 + h) - F(x_0) - BAh\|}{\|h\|} \\ &\leq \frac{\|BE_1(h)\,\|h\|\,\| + \|E_2(k)\,\|k\|\,\|}{\|h\|} \\ &\leq \|B\|\,E_1(h) + \|E_2(k)(\|A\| + \|E_1(h)\|)\|. \end{aligned} \tag{116}$$

Now $\|B\|$ and $\|A\|$ are fixed numbers, and

$$E_1(h) \to 0 \quad \text{as} \quad h \to 0.$$

Furthermore, $k \to 0$ as $h \to 0$ by (113) and the continuity of f at x_0, so

$$E_2(k) \to 0 \quad \text{as} \quad h \to 0.$$

Thus the expression in (116) approaches zero as $h \to 0$. But we have seen that this expression dominates the quotient in (110), so (110) is satisfied, and F is differentiable at x_0 with derivative BA. Since

$$BA = g'(f(x_0))f'(x_0),$$

our proof is complete. ■

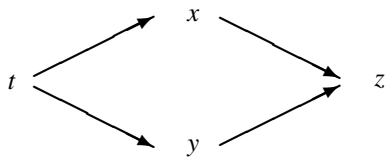
Exercises

12.8.7 If $u : \mathbb{R}^n \to \mathbb{R}$ is differentiable, then we obtained earlier (Example 12.25) the formula

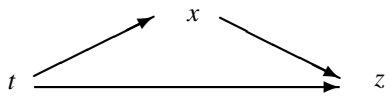
$$d(e^u) = e^u\, du$$

but we were required to assume that e^u is also differentiable. Use Theorem 12.58 to justify that assumption and to check the formula.

12.8.8 Use Theorem 12.58 to obtain an explicit formula for the schema



12.8.9 Use Theorem 12.58 to obtain an explicit formula for the schema



12.8.10 State a theorem that covers a situation of the form

$$F = h \circ g \circ f.$$

SEE NOTE 286

12.8.11 Let

$$\begin{array}{lll} x = r\cos\theta & u = x - y & f(r,\theta) = (x,y) \\ y = r\sin\theta & v = x + y & g(x,y) = (u,v)\ . \end{array}$$

(a) Calculate f', g' and $(g \circ f)'$ directly from the definition of derivative.

(b) Calculate $(g \circ f)'$ from the chain rule.

(c) Calculate the magnification factors for f at $(2,0)$, g at $f((2,0))$, and g at $(2,0)$.

12.8.12 Let

$$\begin{array}{lll} x = re^s & u = y^2 & f(r,s) = (x,y,z) \\ y = se^r & v = x + z & g(x,y,z) = (u,v) \\ z = r & & \end{array}$$

(a) Calculate f', g' and $(g \circ f)'$ directly from the definition.

(b) Calculate $(g \circ f)'$ from the chain rule and compare with the result in (a).

(c) Check that $(g \circ f)'(3,4) = \begin{pmatrix} 32e^6 & 8e^6 \\ e^4 + 1 & 3e^4 \end{pmatrix}$.

12.8.13 Prove the inverse function theorem of Section 12.6.4 using the chain rule of this section.

SEE NOTE 287